

**STUDIES ON ROOT ROT [*Macrophomina phaseolina* (Tassi) Goid.]
OF CHICKPEA (*Cicer arietinum* L.) AND
IT'S MANAGEMENT**

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**BY
PATEL SUJALKUMAR NAGINBHAI
B. Sc. (Agri.)**

**DEPARTMENT OF PLANT PATHOLOGY
B. A. COLLEGE OF AGRICULTURE
ANAND AGRICULTURAL UNIVERSITY
ANAND-388 110
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**STUDIES ON ROOT ROT (*Macrophomina phaseolina* (Tassi) Goid.)
OF CHICKPEA (*Cicer arietinum* L.) AND
IT'S MANAGEMENT**

Name of the student
Patel Sujalkumar N.

Major Guide
Dr. V. P. Gohil

**B. A. COLLEGE OF AGRICULTURE
ANAND AGRICULTURAL UNIVERSITY
ANAND-388 110**

ABSTRACT

Chickpea (*Cicer arietinum* L.) is the third most important grain legume globally grown in over 40 countries. It plays an important role as a grain legume in dry land agriculture in India. It occupies very important position in semi-arid farming system both for human nutrition and restoring the soil fertility. The present investigations on “Studies on Root rot [*Macrophomina phaseolina* (Tassi) Goid.] of chickpea [*Cicer arietinum* L.] and its management” were undertaken at Department of Plant Pathology, B. A. College of Agriculture, Anand Agricultural University, Anand during *Rabi* 2009 – 2011 to explore the possibilities of bioagents, organic manures and chemicals for the management of root rot disease.

The isolation from the roots of freshly infected chickpea plants (GG-2) collected from the farmers fields of Anand districts and

.....*Abstract*

found the association of *Macrophomina* sp. was purified by hyphal tip method and maintained on PDA slants for further investigations.

The pathogenicity test carried out by artificial-soil inoculation method and proved the Koch's Postulates. The symptoms of root rot was found 50-60 days after soil inoculation in chickpea plants from lower to upper leaves become drying from the tip with dark coloured shredded outer sheath of roots. It proved that the isolate was pathogenic to chickpea plants. The reisolated pure culture was identified as *Macrophomina phaseolina* (Tassi) Goidanich (ITCC No. 7810.10).

In the cultural and morphological studies, the colonies of *M. phaseolina* was initially appeared as dirty white mycelium then turn to fluffy white to black with minute dark black sclerotia on PDA medium. The hyphal branch was at right angles with constriction of hyphal branches at their point of origin with closed septum. The microsclerotia were dark black in colour and varied in size with 90-140 µm. The pycnidia were not observed on cultural media.

The variability among the different isolates of *M. phaseolina* was studied to determine the growth parameter and sclerotial formation of each isolates. On 7th day of incubation, the mycelial growth was maximum in all the 4 isolates, the isolates Mp 1 and Mp 2 formed maximum number of sclerotia i.e 75.25 and 68.50, respectively. *M.*

.....Abstract

phaseolina grew best at 30⁰ C temperature showing maximum mycelial growth (89.75 mm) in the isolate Mp 1 *in vitro* conditions.

From the study on effect of solid cultural media, the highest radial growth and excellent sclerotial formation was obtained on PDA (88.25 mm) and proved best followed by Richard's Agar (81.87 mm) for growth of *M. phaseolina*.

Five fungal and 2 bacterial antagonists screened against *M. phaseolina* by dual culture technique *in vitro*. *T. harzianum* and *T. viride* observed as potential antagonistic agents of root rot of chickpea.

Nine fungicides were screened against root rot of chickpea caused by *M. phaseolina in vitro* by poisoned food technique at different concentrations. Significantly cent percent mycelial growth inhibition recorded in bavistin, sixer, cosko, tilt, amistar Top, thirum and vitavex at all the 3 concentrations, while topsin M at 750 ppm recorded 82.22 per cent growth inhibition of the *M. phaseolina*.

Bio-efficacy of 5 herbicides against *M. phaseolina* tested *in vitro* by poisoned food technique at 100, 150 and 200 ppm concentrations. Among the all, alachlor at all the 3 concentrations showed 100 per cent mycelial growth inhibition, while pendimethalin (98.42%), atrazin (33.31%) and 2-4 D (15.92%) also inhibited the growth of *M. phaseolina*.

.....*Abstract*

Experiment conducted in glass house condition with 4 sowing dates and 4 different soil moisture levels against root rot incidence. Disease incidence was higher (72.68%) at water stress conditions and lowest (1.28%) at high moisture conditions.

To manage root rot disease 2 fungal and 1 bacterial antagonists, 2 organic manures and 2 chemicals were evaluated in the field. The combined soil application of *T. viride* @ 5 kg/ha + 500 kg FYM/ha showed the lowest disease incidence (37.27%) and the highest grain yield (3273.6 kg/ha) by controlling 44.43% root rot disease. While *T. harzianum* @ 5 kg/ha + 500 kg FYM/ha showed lower incidence (38.82%) and higher grain yield of 2827.2 kg/ha and controlled 42.12% root rot disease. Whereas, *T. viride* @ 5 kg/ha + 500 kg castor cake/ha gave 41.56% incidence with yield of 2529.6 kg/ha and controlled 38.04% disease.

Dr. V. P. Gohil
Professor
Department of Plant Pathology
B. A. College of Agriculture
Anand Agricultural University,
Anand – 388 110

CERTIFICATE

This is to certify that the thesis entitled “**STUDIES ON ROOT ROT (*Macrophomina phaseolina* (Tassi) Goid.) OF CHICKPEA (*Cicer arietinum* L.) AND IT’S MANAGEMENT**” submitted by **MR. PATEL SUJALKUMAR NAGINBHAI** in partial fulfillment of the requirements for the award of the degree of **MASTER OF SCIENCE (AGRICULTURE)** in the subject of **PLANT PATHOLOGY** by the Anand Agricultural University is a record of bonafide research work carried out by him under my guidance and supervision and the thesis has not previously formed the basis for the award of any degree, diploma or other similar title.

Place: Anand

(V. P. GOHIL)

Date: /06/2011

MAJOR GUIDE

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(Sujal Patel)

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ABBREVIATIONS AND ACRONYMS

%	Per cent
@	At the rate
μ	Micro
μm	Micrometer
°C	Degree Celsius
AAU	Anand Agricultural University
Anon.	Anonymous/Unacknowledged
B. A. C. A.	Bansilal Amritlal College of Agriculture
CD	Critical Difference
cfu	Colony Forming Unit
cv.	Crop Variety
CV	Coefficient of Variation
CzAM	Czapek's Agar Medium
DAS	Days After Sowing
dia.	Diameter
e.g.	For example
et al.	et alii; and worker
EC	Emmursifiable concentration
Fig.	Figure
f. sp.	Forma Species
FYM	Farm Yard Manure
g	Gram
ha	Hectare
HgCl ₂	Mercuric Chloride Acid
hrs	Hours
I.A.R.I.	Indian Agriculture Research Institute
ICRISAT	International Crop Research Institute For Semi-Arid Tropics
i.e.	That is
I.T.C.C.	Indian Type Culture Collection

Kg/cm ²	Kilograms per centimetre square
Kg/ha	Kilograms per hectare
m	Meter
mg	Milligram
ml	Milliliter
mm	Millimeter
min	Minutes
MT	Metric Tone
N	Normal
No.	Number
OMA	Oat Meal Agar
PDA	Potato Dextrose Agar
PDI	Per cent Disease Index
PGI	Per cent Growth Inhibition
pH	Hydrogen ion concentration
ppm	Parts per million
PSAM	Peptone Sucrose Agar Medium
Q	Quintal
RAM	Richard's Agar Medium
SAM	Sabouraud's Agar Medium
SC	Soluble concentration
S. Em.	Standard Error of mean
sp.	Species
Sr. No.	Serial Number
Tr. No.	Treatment Number
t/ha	Ton per hectare
viz.,	Namely
WP	Wettable powder
W/W	Weight by weight
W/V	Weight by volume
ZnSo ₄	Zinc Sulphate

I. INTRODUCTION

The pulses are very important source of protein. Besides, pulse crops keep the soil fertile in respect of nitrogen. On account of these virtues, pulse crops remain an integral part of the sustainable arid-tropics. Chickpea (*Cicer arietinum* L.) is the third most important grain legume globally is grown in more than 40 countries.

Chickpea belongs to the family Leguminosae and believed to be introduced into India from Western Asia. Chickpea is known by other name i.e Gram, Spanish pea, Chestnut bean (English), Poischiche (French), Homos (Arabic), Garbanzo (Spanish) and Chana (Hindi). Mainly two types of chickpea are grown, brown seeded types (Desi) and white seeded called “Kabuli” (Haware *et al.* 1986). It is grown both as irrigated and rainfed crop. However, it’s cultivation as post-rainy season crop is more common.

On an average, chickpea contains 23% protein, 64% carbohydrate, 47% starch, 5% fat, 6% crude fiber, 6% soluble sugar and 3% ash. It also contains various minerals viz., phosphorus (340 mg/100 g), calcium (190 mg/100 g), iron (7 mg/100 g) and zinc (3 mg/100 g). Chickpea is mainly consumed for human diet and only a small proportion is used as animal feed. Germinated seeds are recommended against

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scurvy. Malic and oxalic acid collected from green portions of the chickpea leaves are prescribed for intestinal disorders.

In India, chickpea is grown on 7.58 million hectares with 6.91 million tones of production and 780 kg/ha productivity. In Gujarat, area under chickpea has been 0.25 million hectares with total production of 0.21 million tones and productivity of 870 kg/ha (Anon., 2009). Madhya Pradesh, Rajasthan, Uttar Pradesh, Maharashtra, Karnataka and Gujarat are important states growing chickpea. Panchmahals, Dahod, Ghed area of Junagadh, Bhal areas of Anand and Ahmadabad, Surendra nagar and Bhavnagar are important pockets of chickpea cultivation in Gujarat (Rathod, 2000).

The chickpea crop suffers from many diseases caused by fungi, bacteria, viruses and nematodes. Nene (1996) reported about 67 fungi, 3 bacteria, 22 viruses and 80 nematodes infecting chickpea. Among these Fusarium wilt (*Fusarium oxysporum f. sp. ciceri*), root rot (*Macrophomina phaseolina* (Tassi) Goidanich), Ascochyta blight (*Ascochyta rabiei*), black root rot (*Fusarium solani*), stem rot (*Sclerotinia sclerotiorum*), foot rot (*Operculella padwickii*), rust (*Uromyces ciceri*) and collar rot (*Sclerotium rolfsii*) are of considerable significance (Nene, 1984). In recent years there has been concern about gradual decline in per

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capita availability of pulses. An important reason for this miserable situation in the country is mainly due to the attack of the soil borne pathogens (Eswaran and Mishra, 2004).

Macrophomina phaseolina is a most important and very serious soil borne pathogen. Production of both pycnidia and sclerotia is an important characteristic. It belongs to the family Sphaeropsidaceae of the order Sphaeropsidales under the class Deuteromycetes. The pycnidial stage was initially named as *M. phaseolina* by Tassi in 1901 and *M. phaseoli* by Maublanc in 1905. In 1927, Ashby maintained the name *M. phaseoli*, while Goidanich in 1947 proposed *M. phaseolina* (Mbaye, 2007). In India (Pusa), the first report was made on macrophomina root rot of Chickpea by Mitra in 1931.

M. phaseolina (Tassi) Goid. [*R. bataticola* (Tabu.) Butler] incites disease in a wide range of hosts especially under high temperature and drought stress conditions (Singh and Nene, 1990). The pathogen is soil borne and survives in soil for several years (Taya *et al.*, 1990). The disease may cause losses to the tune of 70.8% at full podding stage and 48.9% at pre harvest stage of the crop (Ahmad and Mohammad, 1986).

Considering importance of the crop, its area under cultivation and losses due to chickpea root rot (*M. phaseolina*) disease,

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the attempts were made to the present investigations for management of root rot disease of chickpea with following aspects.

Research aspects:

- 1.** Isolation, pathogenicity and confirmation of the pathogen
- 2.** Studies on variability among the chickpea isolates of
M. phaseolina
- 3.** Effect of solid culture media on the growth of *M. phaseolina*
in vitro
- 4.** Studies on antagonism between root rot pathogen and antagonists
in vitro
- 5.** Bio-efficacy of fungicides against *M. phaseolina in vitro*
- 6.** Studies on effect of herbicides on the growth of *M. phaseolina*
in vitro
- 7.** Studies on influence of sowing dates, soil moisture levels and incidence of chickpea root rot in pot conditions
- 8.** Management of Root rot of chickpea in field conditions

II. REVIEW OF LITERATURE

Available literature on various aspects related to present investigation is reviewed here under.

History:

The genus *Rhizoctonia* was erected by deCandole (1815). According to Garrett's (1951) ecological grouping of fungi, it is placed under soil inhabitant. *Macrophomina phaseolina* is reported as the pycnidial stage of *R. bataticola* (Ashby, 1927 and McRae, 1929). Goidanich 1947 examined the original material of *M. phaseoli* and renamed it as *M. phaseolina* (Tassi) Goid. The fungus is characterized by branching of hyphae at right angles with a constriction of hyphal branches at their point of origin. The microsclerotia are black in colour. It belongs to the family Sphaeropsidaceae, order Sphaeropsidales, class Deuteromycetes and phylum Ascomycota.

Symptomatology:

Singh *et al.* (1990) reported that the disease occurs after flowering stage and its incidence increases with increasing growth stage of crop up to the harvesting stage. Whole plant withers without showing conspicuous flaccidity symptom. Leaves along with petioles, branches

.....*Review of Literature*

and stem become straw – coloured due to drying. Upon uprooting, affected plants indicate darkening of roots and shredding of bark. Minute black colored sclerotial bodies can be visible inside the shredded bark as well as in pith portion. Monga *et al.* (2007) reported that cotton root rot (*R. bataticola*) was characterized by sudden and complete wilting of the plants. The affected plants can easily be pulled out of ground. The bark of roots developed cracks and gives yellowish appearance in diseased plants. Naik and Gohil (2010) also observed such symptoms in a farmers fields in the area of Anand and Ahmadabad districts at flowering and full podding stages.

Losses:

Losses in yield due to dry root rot caused by *Rhizoctonia bataticola* of chickpea at full podding stage have been estimated to the extent of 70.8% and at pre harvest stage 48.9%. The per cent reduction in 1000 seed weight was 18.4% at pre harvest stage and 33.9% at full podding stage in Bihar (Ahmad and Mohammad, 1986).

Factors affecting incidence:

Soil moisture:

Soil moisture plays an important role in the development of disease by soil inhabitants. Haware (1980) stated that the high temperature and

.....*Review of Literature*

water stress conditions were important factors in disease development due to *Rhizoctonia*. *M. phaseolina* attacked maximum under moisture stress condition and decreased with increasing soil moisture level (Sandhu and Singh, 1999). *M. phaseolina* infection was maximum at 25% and least at 100% soil moisture (Patel and Anahosur, 2001). As the soil moisture per cent increased, the intensity of disease decreased progressively (Mukherjee, 2003).

Temperature:

Temperature is an important factor for the growth of the fungus. The best growth of the fungus was at 30⁰ and 35⁰ C temperature. The sclerotial formation was also maximum at this temperature (Ratnoo and Bhatnagar, 1991). The higher temperature range of 25-35⁰C supported the growth of *M. phaseolina* and 25⁰C temperature favored sporulation of the fungus (Sharma *et al.* 2004).

pH:

The growth of *M. phaseolina* increased gradually with increasing pH up to 7.0 (Ratnoo and Bhatnagar, 1991). The growth of *M. phaseolina* was maximum in the pH range of 6.5-7.0 (Sharma *et al.* 2004).

1. Isolation, pathogenicity and confirmation of the pathogen

1.1 Isolation of the pathogen:

Singh and Nene (1990) isolated pure cultures of *R. bataticola* from infected chickpea, pigeon pea, groundnut and sorghum

.....*Review of Literature*

and reported that chickpea isolate was more pathogenic to safflower, sorghum, bajara and pigeon pea.

Baird *et al.* (1995) isolated three different fungi including *Fusarium* sp., *R. solani* (AG-4) and binucleate *Rhizoctonia* sp. (AG-5) from diseased plants of cotton in Tifton and found that the AG-4 isolate was less damaging than the binucleate *Rhizoctonia* isolate. When the experiment was repeated, similar results were observed.

Pratt *et al.* (1998) reported the tissue (root) isolation method of *M. phaseolina* from alfalfa and white clover from the North America.

Demirci *et al.* (1999) proved the pathogenicity of *M. phaseolina* from plants and seeds of chickpea under controlled environmental condition.

Patel and Anahosur (2000) studied the association of soil borne fungi in chickpea plants infected with wilt. Isolations were made at 10 days interval from 10th day of sowing upto 90th day and found association of *M. phaseolina*, *Fusarium* spp. and *Sclerotium rolfsii* Sacc.

Singh and Sirohi (2003) isolated *R. bataticola* from all the diseased parts of chickpea plants.

Katariya *et al.* (2007) isolated nine different isolates of *R. bataticola* from dry root rot of chickpea (cv. C-235) diseased plants and found that the *R. bataticola* was most pathogenic in sterilized and unsterilized sandy loam soil.

Naik and Gohil (2010) isolated *M. phaseolina* from infected roots of chickpea at flowering and podding stage for the first time in middle Gujarat.

.....*Review of Literature*

1.2 Pathogenicity and conformation of the pathogen (Koch's Postulate):

Grover and Sakhuja (1981) proved the pathogenicity of *R. bataticola* in mung bean using mycelial suspension prepared from 8-10 days old cultures @ 250 mg fresh weight per pot containing 1 kg sterilized soil and kept moist for 2-3 days for stabilizing the pathogen and reported that the spraying of the seedlings or detached leaves with 4-8 days old mycelial suspension of *R. bataticola* growing on Czapek's nutrient solution at 30⁰ C temperature was most effective in producing typical lesions within 36-48 hrs.

Goyal and Jain (1998) developed two different methods viz; seed inoculation and soil inoculation technique of *M. Phaseolina* for pathogenicity test in black gram. They found that in soil inoculation highest pre (48.10%) and post emergence (37.33%) mortality was observed with *M. phaseolina* and *Fusarium* sp. respectively, as compared to seed inoculation.

Pratt *et al.* (1998) reported *M. phaseolina* on alfalfa and white clover in North America. They proved pathogenicity by inoculating tissues of mature plants with tooth pick pieces method and reported that high solute (NaCl) concentrations reduced the synthesis of mycelium pigments and the size and shape of microsclerotia.

Garcia *et al.* (2003) proved pathogenicity of *M. phaseolina* on common bean *in vitro* conditions.

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Singh and Sirohi (2003) proved pathogenicity of chickpea isolates *R. solani* and *Fusarium* spp. on susceptible chickpea cultivar JG-62 by using pot inoculation method developed by Nene and Haware (1980).

Jaiman and Jain (2008) proved pathogenicity of *M. phaseolina* cause of blight and root rot disease in cluster bean by seed inoculation.

2. Studies on variability among the chickpea isolates of *M. phaseolina*

Patil *et al.* (2001) studied the variability in different six isolates of *M. phaseolina* obtained from major *Rabi* sorghum growing areas of Maharashtra (Sholapur) and Karnataka (Dharwad and Bijapur) in India in respect of cultural characters showed that each of the isolates was distinct with one another. Growth of Sholapur isolate revealed more mycelia weight (13.3mg) than Bijapur (12.9mg) and Dharwad (8.3mg).

Sharma *et al.* (2004) reported *M. phaseolina* from pearl millet, sesame, horsegram and mothbean crops and reported variation in size, margin and colour of the fungal colonies and also differences in the size of sclerotia.

Kar and Mohantray (2005) reported three isolates of *M. phaseolina* obtained from the naturally infected roots, leaves and seeds of mung bean from farmers' fields and marked variation in their

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morphological and cultural characteristics. The isolates also varied significantly in their pathogenic potential on different plant parts, leaf isolate being more aggressive than root and seed isolates.

Ratnoo and Bhatnager (1991) studied the growth and sporulation of *M. phaseolina* at different temperature viz; 15, 20, 25, 30, 35 and 40°C for 72 hr. *M. phaseolina* isolated from cowpea grew best and formed maximum sclerotia at 30 and 35°C temperature. Total mortality of lentil plants due to *M. phaseolina* was found maximum at 30°C temperature.

Sandhu and Singh (1999) studied the effect of temperature at 15, 20, 25, 30, 35 and 40°C and reported 30°C as an optimum temperature for the growth of the *M. phaseolina* of cowpea in pot condition.

Sharma *et al.* (2004) studied the effect of temperature and reported that the higher temperature range of 25-30°C favoured the growth of *M. phaseolina in vitro*.

Jha and Sharma (2005) studied the effect of temperature from 20 to 40°C and reported that the optimum temperature for *R. bataticola* was 30-35°C both for mycelial growth and sclerotial formation. At 40°C temperature, very poor mycelial growth and no sclerotial formation was observed in the study.

3. Effect of solid culture media on the growth of *M. phaseolina in vitro*

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Grover and Sakhuja (1981) reported that fungus developed sclerotia on Czapek's agar within 72 hrs and pycnidial stage was never observed in culture or on dried leaves *in vitro*.

Kulkarni *et al.* (1992) compared different culture media viz; Czapek's agar, Cotton root extract agar, Oat meal agar, PDA, Richard's agar, Sabouraud agar and Tyrosine agar for the growth and sclerotial formation in *R. bataticola* and reported that growth of *R. bataticola* was significantly superior on Cotton root extract agar, PDA, Richard's agar and Sabouraud's agar as compared to other media. Sclerotial production was excellent on Richard's agar, PDA, Sabouraud agar and Czapek's agar media.

Sobti and Sharma (1992) studied the five culture media viz; PDA, Oat meal agar, Czapek's agar and Peptone sucrose agar media for mycelia growth and sclerotial production of *M. phaseolina*.

Suriachandraselvan and Seetharaman (2003) studied the mycelia growth and sclerotial production on PDA, Oat meal agar, Richard's agar, Czapek's agar and Pepton sucrose agar media and found PDA medium the most effective (89.6 mm) followed by Richard's agar (76.7 mm) and Oat meal agar (75.1 mm) medium with maximum sclerotial production.

Shekhar *et al.* (2006) studied 7 isolates of *M. phaseolina* on the basis of colony colour, growth rate and sclerotial production and

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reported that sclerotial size and number had positive correlation with virulence of charcoal rot of maize.

4. Studies on antagonism between root rot pathogen and antagonists *in vitro*

Selvarajan and Jeyarajan (1996) studied 8 fungal and 3 bacterial antagonists against root rot of chickpea caused by *M. phaseolina* and *Fusarium solani* and reported that *Trichoderma* spp., *Laetisaria arvalis*, *B. subtilis* and *P. fluorescens* showed inhibition zones with reduced sclerotial size, germination and germ tube number of *M. phaseolina in vitro*.

Rajeshwari *et al.* (1998) studied the antagonists *in vitro* by dual culture technique and reported that the maximum inhibition of *M. phaseolina* was obtained with *T. harzianum* (69.28%) followed by *Trichoderma viride* (64.94%), *Gliocladium virens* (64.84%) and least with *T. koningii* (49.06%).

Ahamad and Shrivastava (2000) reported that *T. viride*, *B. subtilis* and *P. fluorescens* were effective against dry root rot of chickpea (*R. bataticola*) by dual culture technique.

Sindhani *et al.* (2002) studied the antagonistic activity of *T. viride*, *T. harzianum*, *A. flavus*, *A. niger*, *A. oryzae*, *Penicillium citrinum*, *B. subtilis* and *P. fluorescens* against *R. bataticola in vitro* and

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reported that all the antagonists inhibited the mycelial growth and sclerotial formation.

Desai and Kulkarni (2002) found that *P. fluorescens* most effective against *M. phaseolina*. All the *Trichoderma* spp. recorded more than 50 and 25% hyphal and sclerotial inhibition, respectively.

Kaswate *et al.* (2003) studied 3 antagonists against *M. phaseolina* from different crops including chickpea and reported that *T. viride* gave 100% inhibition followed by *B. subtilis* (87.41-92.89%) and *P. fluorescens* (73.98-78.94%) *in vitro*.

Eswaran and Mishra (2004) studied the antagonistic effect of 3 *Aspergillus* spp. against *M. phaseolina in vitro* and reported the inhibition of *A. terreus* (31.8 mm), *A. nidulans* (58.4 mm) and *A. clavatus* (42.1 mm) by dual culture technique.

Suriachandraselvan *et al.* (2004) screened 10 fungal antagonists by dual culture technique. They reported that *T. viride* and *T. harzianum* significantly inhibited the mycelial growth of *M. phaseolina*.

Kumar *et al.* (2007) studied different isolates of *P. fluorescens* against *M. phaseolina* by dual culture technique and reported maximum inhibition zone (7.2-10.8 mm) of the test pathogen.

Bandopadhyay *et al.* (2008) studied the antagonistic effect by 8 different fungal biocontrol agents and reported that *T. viride*, *G. virens*, *A. niger* and *A. fumigatus* were most effective against *M. phaseolina* of cortical stem and root rot diseases in jute by dual culture technique.

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Choudhary *et al.* (2010) tested 4 bioagents viz; *T. viride*, *T. harzianum*, *A. versicolor* and *B. firmus* against dry root rot of mung bean by dual culture technique and found *T. viride*, *T. harzianum* and *A. versicolor* effective in inhibiting the mycelial growth of *M. phaseolina* (61.65%).

Naik *et al.* (2010) screened 7 bioagents against root rot of chickpea by dual culture technique *in vitro* and reported that *T. harzianum* was the most effective against inhibiting the growth of *M. phaseolina* while, *T. viride* and *T. virens* were found second best against the pathogen causing chickpea root rot disease.

5. Bio-efficacy of fungicides against *M. phaseolina* *in vitro*

Majumdar and Som (1987) studied 3 systemic and 3 non systemic fungicides by poisoned food technique and reported that Calixin was found as the most potent and broad spectrum fungicide next in order to Bavistin. Among the 3 non systemic fungicides, Hexaferb was most effective followed by Foltaf for inhibiting mycelial growth of test pathogen.

Prajapati *et al.* (2002) studied the efficacy of 11 fungicides against *M. phaseolina* and reported that Carbendazim, Carbendazim + Thirum, Carboxin and Topsin-M completely inhibited the fungal growth of *M. phaseolina* *in vitro*.

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Lambhate *et al.* (2002) studied the efficacy of 6 fungicides against root rot caused by *M. phaseolina* in cotton and reported that Bavistin, Ridomil M Z-72 and Topsin-M @ 0.1, 0.2 and 0.3% showed complete inhibition of mycelial growth of the fungus, respectively by poison food technique.

Chattopadhyay and Sastry (2002) screened the fungicidal effect by poison food technique and reported that Carbendazim @ 25 ppm and Benomyl and Thirum at 100 ppm were found most effective by inhibiting 100 % mycelial growth of *M. phaseolina*.

Sudhakar *et al.* (2002) studied 12 agrochemicals against *M. phaseolina* causing charcoal rot of maize *in vitro* and reported that Tridimefon, SAAF and Hexaconazole gave 100% inhibition of mycelial growth followed by Companion (92.2%) and Carboxin (81.2%).

Jha and Sharma (2006a) screened 5 different fungicides at 10, 50 and 100 ppm concentrations against *R. bataticola* by poison food technique and reported that Bavistin was most effective and checked growth of all isolates at all the concentrations.

Suryawanshi *et al.* (2008) studied the efficacy of 7 fungicides by poison food technique against *M. phaseolina* and reported that Carbendazim @ 0.05%, Mancozeb @ 0.2% and Thirum @ 0.3% were the most effective fungicides with maximum radial growth inhibition (94.18%).

6. Studies on effect of herbicides on the growth of *M. phaseolina*

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Iqbal *et al.* (1996) studied the efficacy of two different herbicides viz; Methabenzthiazuron and Chlorotoluron on *M. phaseolina* by dual culture technique and found both the herbicides effective.

Lima *et al.* (1997) studied the effect of Alachlor and Benomyl on the mycelial growth and germination of sclerotia of *M. phaseolina* and found that Alachlor @ 1000 and 2000 ppm inhibited the mycelial growth and found at par with Benomyl @ 100 and 300 ppm, respectively. Alachlor @ 1000 and 2000 ppm and Benomyl @ 300 ppm inhibited the fungal growth and sclerotial germination.

Jha and Sharma (2006b) screened 6 different herbicides against *R. bataticola* by poison food technique and reported that Alachlor @ 100 and 200 ppm inhibited the mycelium growth.

7. Studies on influence of sowing dates, soil moisture levels and incidence of chickpea root rot in pot conditions

Husain and Ghaffar (1995) reported that colonization of *M. phaseolina* on chickpea root rot was higher when plants were moisture stressed compared with unstressed plant.

Patel and Anahosur (2001) evaluated different 3 dates of sowing i.e October 1st and 15th and November 1st with four available moisture per cent and observed the presence of *M. phaseolina* and reported that lower moisture (25%) gave higher disease incidence

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(47.80%) and higher moisture (100%) gave lower disease incidence (1.28%).

8. Management of root rot of chickpea in field conditions

8.1 Bioagents:

Wells *et al.* (1972) first demonstrated the efficacy of *T. harzianum* against *Rhizoctonia spp.* in the field.

Singh and Sindhan (1998) reported that among the treatments viz. *T. viride* and *T. harzianum* as seed treatment, soil application and soil application + seed treatment, the treatment of soil application + seed treatment were found best followed by soil application and seed treatment alone in reducing the dry root rot incidence.

Ahamad and Srivastava (2000) reported that soil inoculation with *B. subtilis* (10^9 cfu/ml) gave 29.56% and seed treatment with *T. viride* (10^8 cfu/ml) gave 32.50% disease control of dry root rot of chickpea.

Dubey (2002) reported that seed treatment with Vitavax (0.1%) + *T. viride* (0.2%) resulted into maximum seed germination (88.4%) and minimum disease incidence (4%) with highest pod yield (2002 g/plot) in collar rot of french bean caused by *R. bataticola*.

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Sethuraman *et al.* (2003) reported that seed treatment + soil application of *T. viride* showed significantly less incidence (4.71%) of root rot (*M. phaseolina*) disease in blackgram.

Mani and Hepziba (2003) reported that seed treatment with *T. viride* combined with the soil application of neem cake 250 kg/ha showed 19.18% disease incidence followed by *T. harzianum* + neem cake (29.68%).

Suriachandraselvan *et al.* (2004) reported that seed treatment with talc based formulation of *T. viride* and *T. harzianum* @ 4 g/kg seed significantly reduced the disease incidence (22-27%) as compared to control (67%).

Sajeena *et al.* (2004) studied the seed treatment and soil application with *T. viride* and *P. fluorescence* and reported increased germination and reduced seed mortality.

Gaur *et al.* (2005a) reported that soil application of 10-15 days pre-inoculated wheat bran based *Trichoderma* @ 10 kg/ha in 200 kg of FYM gave significantly lowest mortality (2.4%) due to root rot (*R. bataticola*) followed by seed treatment with *T. harzianum* @ 10 g/kg seed in combination with soil application of ZnSo₄ @ 25 kg/ha in cotton.

Gaur *et al.* (2005b) reported that soil application of 10 to 15 days pre-inoculated *T. harzianum* @ 10 kg/ha in 200 kg of FYM was

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found effective followed by soil application of ZnSO_4 @ 25 kg/ha combined with seed treatment of *T. harzianum* @ 10 g/kg seed in root rot of chickpea.

Chavan (2006) reported that combined application of *T. harzianum* with FYM was more effective for the root rot disease control (68.39%) with higher germination and cotton yield.

Jaiman *et al.* (2009) studied five different bioagents against root rot of cluster bean and reported that seed treatment with *T. viride* @ 4 g/kg showed reduction in disease incidence (19%).

Choudhry *et al.* (2010) reported that the combination of soil application of biocontrol agents *T. viride* and *T. harzianum* reduced 23%, while *T. harzianum* and *A. versicolor* reduced 31% root rot incidence.

Naik *et al.* (2010) reported that *T. viride* @ 2.5 kg/ha with FYM @ 500 kg/ha (SA) was found most effective against chickpea root rot (20.45%) with yield of 1327.25 kg/ha by controlling 63.13% disease.

8.2 Chemical:

Singh *et al.* (1986) reported Carbendazim @ 0.05% the most effective in reducing Macrophomina blight disease (30.2% DI) with highest grain yield (6.87q/ha) in mungbean.

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Bagri *et al.* (2004) reported that Thirum gave the highest seed germination (89%) and minimum pre and post emergence mortality of chickpea seed infected by *M. phaseolina*.

Sharma and Gupta (2004) reported that seed treatment with Carbendazim @ 1 g/kg seed showed minimum incidence (19%) and gave highest yield (1299 kg/ha) in dry root rot of chickpea.

Rathore (2006) reported that seed treatment with Carbendazim (2 g/kg) recorded minimum incidence (14.1%) with maximum seed (132 kg/ha) and fodder (10.14 q/ha) yield in green gram.

Gore *et al.* (2008) reported that the maximum per cent disease control (10.76%) was achieved by Carbendazim @ 0.05% in mung bean blight (*M. phaseolina*).

Suryawanshi *et al.* (2008) studied 8 different fungicides against mung bean blight caused by *M. phaseolina in vivo* and reported that Carbendazim @ 0.05% gave least disease incidence (10.67%) and highest disease control (58.45%) with 429.67 kg/ha yield.

Jaiman and Jain (2008) studied the efficacy of fungicide against *M. phaseolina* and reported that seed treatment with Carbendazim @ 2 g/kg showed lowest disease incidence and maximum grain yield in root rot of cluster bean.

Jaiman *et al.* (2009) studied 6 fungicides against root rot of cluster bean and found that Carbendazim @ 2g/kg of seed showed minimum pre and post emergence seed rot disease incidence (10%) and maximum grain yield (859 kg/ha).

8.3 Oil cake:

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Rajani (1998) reported the application of neem cake @ 20 q/ha effective for the management of root rot (*M. phaseolina*) disease in castor followed by mustard and castor cake.

Jaiman *et al.* (2009) studied the different 6 soil amendments and reported that the application of neem cake @ 25q/ha gave least disease incidence (18%) as compare to other treatments.

Naik *et al.* (2010) reported that *T. viride* @ 2.5 kg/ha with castor cake @ 500 kg/ha (SA) showed lower disease incidence (24.15%) and controlled 56.47% root rot disease with the yield of 1195.00 kg/ha in chickpea.

III. MATERIALS AND METHODS

All the experiments pertaining to the present studies were conducted at the Department of Plant Pathology, B. A. College of Agriculture, Anand Agricultural University, Anand during 2009-2010.

3.1 General Information:

3.1.1 Location

Anand Agriculture University, Anand, where the present investigations were under taken is located on 22° -35' North latitude and 72° -55' east latitude and has an elevation of 45 meters above the mean sea level.

3.1.2 Climate:

Climate is generally warm and moderately humid. Monsoon commences by the middle of June and ends in middles of September. Winter is fairly cool and dry, while summer is generally hot where temperature ranges from 30 - 45⁰ C. An average rainfall of the tract is 762 mm.

3.1.3 Laboratory procedures:

3.1.3.1 Cleaning of glassware:

Standard qualities of glassware were used during all the laboratory experimentation. The glasswares were immersed for 24 hours

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in cleaning solution of chromic acid (60g of potassium dichromate dissolved in 300 ml of distilled water and 40 ml concentrated sulphuric acid added with constant stirring) and were washed with the detergent powder followed by tap water and finally rinsed with sterilized distilled water and dried before use.

3.1.3.2 Sterilization of Glassware:

The cleaned glasswares were sterilized at 180°C temperature for 2 hours in hot air oven. Culture media and water were sterilized in an autoclave at 121°C temperature at 1.1 kg/cm² pressure for 20 minutes.

3.1.3.3 Sterilization of soil and earthen pots:

Sandy loam soil mixed with Farm Yard Manure (FYM) in 4:1 proportion and the mixture was steam sterilized at 1.1 kg/cm² pressure for consequent 2 hours for two days and used in the surface sterilized earthen pots for the experimentation. The earthen pots of 15 cm diameter were washed thoroughly with tap water and disinfected by dipping in 4 % formaldehyde (Formalin 40%) solution for 2 minutes and rinsed with sterilized water prior to use.

3.2 Isolation, pathogenicity and confirmation of the pathogen

3.2.1 Isolation of pathogen:

Fresh infected roots of chickpea plants showing typical, well

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developed dark brown to black discolouration were used to isolate the pathogen from the infected parts. The infected roots were cut into small pieces in such a way that each piece comprised of infected as well as healthy tissues. The pieces were surface sterilized with 0.1 per cent mercuric chloride (HgCl₂) solution for 30 seconds followed by three washing with sterilized distilled water and then transferred aseptically using laminar air flow system on sterile Petri plates containing 20 ml Potato Dextrose Agar (PDA) medium and incubated at 28±2°C temperature. After 3-4 days the hyphal tips of the growing fungi were aseptically cut and transferred to PDA slants. After proper growth on the slants, the fungi were identified and stored at 5°C temperature as stock culture.

3.2.2 Pathogenicity

Pathogenicity of the isolate was tested in sterilized soil on chickpea variety 'Gujarat Gram-2'. Inoculum of isolate was multiplied on sterilized sand maize flour medium (partially broken maize grains 10 g, sand 10 g and 20 ml distilled water in 250 ml kept in Erlenmeyer flask). The flask containing the sterilized medium was inoculated with mycelial disc of *Macrophomina* sp. (5 mm dia.) and incubated at 28±2°C for 15 days. This inoculum was used for soil inoculation and added in

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disinfected earthen pots maintained in triplicates. Seed samples were surface sterilized with mercuric chloride (HgCl₂ 0.1%) for 1 min and 5 seeds of GG-2 were sown in each pot. Observations on disease development were recorded after seed germination to harvest.

3.2.3 Confirmation:

The morphological characters viz., type of mycelial growth and sclerotial formation were studied under low power magnification from 10 days old pure culture of *Macrophomina* sp. and were compared with those given in literature. The photomicrograph of sclerotia was also taken. A pure culture obtained was sent to Indian Type Culture Collection, Identification/supply Services, Division of Plant Pathology, IARI, New Delhi -110 012 for identification.

3.3 Studies on variability among the chickpea isolates of *M. phaseolina*

The root rot diseased samples were collected from different locations i.e. Arnej of Ahmedabad district, Dahod of Panchmahal district and Anand and Tarapur from Anand district. Isolation was done on PDA and purified by hyphal tip method. Ten days old pure culture of each isolate was used for the study of cultural, morphological and physiological characteristics. For colony characteristics, 3 PDA plates per

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isolates were inoculated with 5 mm mycelial discs of each individual isolates and incubated at $28 \pm 2^{\circ}$ C. The fungul growth of the colony, its diameter, type of growth and number of sclerotia were recorded on 1st, 5th and 7th day after inoculation. Variability in different isolates of *M. phaseolina* was studied at different temperature ranging from 15-35^o C.

3.4 Effect of solid culture media on the growth of *M. phaseolina* in vitro

With a view to find out the best medium for the growth and sclerotial formation of fungus, six media in solid states were compared.

The compositions of various media used are as under.

1. Czapek's Agar Medium (CzAM)

Sodium nitrate (NaNO ₃)	:	2.0 g
Dipotassium hydrogen orthophosphate (K ₂ HPO ₄)	:	1.00 g
Magnesium sulphate (MgSO ₄ .7H ₂ O)	:	0.50 g
Potassium chloride (KCl)	:	0.50 g
Ferrous sulphate (FeSO ₄ .7H ₂ O)	:	0.01 g
Sucrose (C ₆ H ₂₂ O ₁₁)	:	3.00 g

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Agar agar	:	20.00 g
Distilled water	:	1000.00 ml

2. Oat Meal Agar (OMA)

Oats	:	100.00 g
Agar agar	:	20.00 g
Distilled water	:	1000.00 ml

3. Peptone Sucrose Agar Medium (PSAM)

Sodium glutamate	:	1.0 g
Peptone	:	10.0 g
Sucrose	:	10.0 g
Agar agar	:	17.0 g
Water	:	1000 ml

4. Potato Dextrose Agar (PDA)

Peeled potato	:	200.00 g
Dextrose (C ₆ H ₁₂ O ₆)	:	20.00 g

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Agar agar	:	20.00 g
Distilled water	:	1000.00 ml

5. Richard's Agar Medium (RAM)

Potassium nitrate (KNO_3)	:	10.00 g
Potassium dihydrogen orthophosphate (KH_2PO_4)	:	5.00 g
Magnesium sulphate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$)	:	2.5 g
Ferric chloride ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$)	:	0.02 g
Sucrose ($\text{C}_6\text{H}_{12}\text{O}_{11}$)	:	50.00 g
Agar agar	:	20.00 g
Distilled water	:	1000.00 ml

6. Sabouraud Agar Medium (SAM)

Peptone	:	10.0 g
Dextrose	:	40.0 g
Agar agar	:	15.0 g

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Water : 1000 ml

The cultural study was conducted on PDA in Petri plates (90 mm). Sterilized media (20 ml) was poured into each sterilized Petri plate. The plates were inoculated with fungal disc of 5 mm size cut from periphery of 10 days old pure culture of *M. phaseolina* and incubated at $28\pm 2^{\circ}\text{C}$. Four replications were maintained for each culture medium. Colony diameter was measured diagonally after 3 days of inoculation. Mycelial growth and number of sclerotia and size were recorded after 7 days of incubation.

Observations recorded:

1. Mycelial growth (mm)
2. Sclerotial formation

Table 3.1 Criteria used to record the sclerotial formation of *M. phaseolina*

Sr. No.	Particulars	Sclerotial production/ microscopic field	Sign indication	3.5
1	No sclerotial formation	Nil	-	
2	Poor sclerotial formation	1-20	+	
3	Moderate sclerotial formation	21-30	++	
4	Good sclerotial formation	31-50	+++	
5	Excellent sclerotial formation	>50	++++	

Studies on antagonism between root rot pathogen and antagonists *in vitro*

Antagonism study was carried out by dual culture technique

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using PDA medium. The twenty ml of PDA medium was poured in the Petri plates and was inoculated with the antagonists and test pathogen (5 mm disc of 10 days old culture) opposite to each other near the periphery of Petri plate. The medium incubated with the pathogen alone served as control. The Petri plates were incubated at $28\pm 2^{\circ}\text{C}$ temperature. After 5 days of incubation the growth of the test pathogen was measured. The growth of the test fungi was compared with control. The per cent growth inhibition was calculated by using the formula (Vincent, 1927).

$$\text{Per cent growth inhibition (PGI)} = \frac{C - T}{C} \times 100$$

Where,

C= Average diameter of mycelial colony in control treatment (mm)

T= Average diameter of mycelial colony in treated set (mm)

Table 3.2: Bioagents tested against *M. phaseolina* by Dual culture technique

Sr. No.	Treatments
1	<i>Aspergillus flavus</i>
2	<i>Aspergillus niger</i>
3	<i>Bacillus subtilis</i>
4	<i>Pseudomonas fluorescens</i>
5	<i>Trichoderma harzianum</i>
6	<i>Trichoderma virens</i>
7	<i>Trichoderma viride</i>

3.6 Bio-efficacy of fungicides against *M. phaseolina* in vitro

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The selected fungi-toxicants were screened @ a.i concentration *in vitro*. PDA medium (20 ml) and the fungi-toxicants were thoroughly mixed and poured into sterilized Petri plates (poison food technique). The Petri plates were inoculated by placing 5 mm mycelial disc from ten days old pure culture of *M. phaseolina* with 4 replications and incubated 28±2°C till the radial growth of the colony touch the periphery in control treatment. The observations on linear growth of fungus were recorded at 24 hours interval till the entire plate in control was completely covered with mycelium. The per cent growth inhibition (PGI) of the pathogen over control was worked out by using formula (Vincent, 1927).

$$\text{Per cent growth inhibition (PGI)} = \frac{C - T}{C} \times 100$$

Where,

C= Average diameter of mycelial colony in control treatment (mm)

T= Average diameter of mycelial colony in treated set (mm)

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Table 3.3: Concentrations of fungicides with active ingredients

Sr. No.	Trade name	Technical name	Concentration (ppm)		
1	Bavistin (50%WP)	Carbendazim	250	500	750
2	Sixer (75% WP)	Carbendazim 12% + Mancozeb 63%	500	750	1000
3	Cosko (75%SD)	Carboxin 37.5% + Thiram 37.5%	300	500	700
4	Tilt (25% EC)	Propiconazole	25	50	75
5	Amistar Top (0.125 %)	Azoxystrobin 18.2% + Difenconazole 11.4%	50	100	150
6	Devithiram (75%WP)	Thirum	500	750	1000
7	Topsin M (70%WP)	Thiophenate methyl	250	500	750
8	Vitavax (75% WP)	Carboxin	250	500	750
9	Amistar (23 SC)	Azoxystrobin	250	500	750
10	Control	-	-	-	-

3.7 Studies on effect of herbicides on the growth of *M.*

phaseolina in vitro

Herbicides were incorporated aseptically in sterilized PDA medium at different concentrations and poured (20 ml) aseptically in

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sterilized Petri plates in triplicate for each treatment. Five mm mycelial discs of the isolates of *M. phaseolina* cut from the margin of seven days old cultures were aseptically placed centrally in each of the Petri plates separately. Inoculated plates were incubated at $28 \pm 2^\circ\text{C}$ temperature. The observations on linear growth of fungus were recorded at 24 hr interval till the entire plate in control was completely covered with mycelium. Per cent growth inhibition of the fungal colony was recorded and calculated by the following formula given by (Vincent, 1927).

$$\text{Per cent growth inhibition (PGI)} = \frac{C - T}{C} \times 100$$

Where,

C= Average diameter of mycelial colony in control treatment (mm)

T= Average diameter of mycelial colony in treated set (mm)

Table 3.4: Concentrations of herbicides with active ingredients

Sr. No.	Trade name*	Technical name
1	Lasso 50 EC	Alachlor
2	Atrataf 50% WP	Atrazin
3	Sencor 70 EC	Metribuzin
4	Stomp 30 EC	Pendimethalin
5	Weedmar 80% WP	2,4 – D
6	Control	-

*Concentrations for all treatments are 100, 150, 200 ppm

3.8 Studies on influence of sowing dates, soil moisture levels

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and incidence of chickpea root rot in pot conditions

The experiment was conducted with 4 dates of sowing i.e 15th October, 1st, 15th November and 1st December with 4 available soil moisture levels i.e 25, 50, 75 and 100 %. The inoculum of *M. phaseolina* was multiplied on Sand maize meal medium for 20 days. The surface sterilized earthen pots of 15 cm diameter were filled with sterilized soil. 20 days old well multiplied inoculum of *M. phaseolina* also incorporated in the sterilized soil @ 5% w/w in each pot prior to 2 days of sowing. The surface sterilized chickpea seeds were sown in each pot and 5 plants were maintained in each pot till to harvest. The treatment was replicated 5 times. Soil moisture was maintained by addition of required amount of water as per treatments periodically at 3 days interval. The disease incidence was recorded from flowering to pod stage and harvest stage by the following formula.

$$\text{Per cent Disease Incidence} = \frac{\text{Number of diseased plant}}{\text{Total Number of observed plants}} \times 100$$

3.9 Management of Root rot of chickpea in field conditions

Considering the importance of root rot disease, an experiment was conducted to study the efficacy of various bioagents, oil

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cakes and fungicides as seed treatment as well as soil application for the management of disease during *Rabi* season.

The experiment was carried out at Agronomy Farm, B. A. College of Agriculture, Anand Agricultural University, Anand during *Rabi* 2010-11 in Randomized block design (RBD) along with 10 treatments and 3 replications. All the recommended agronomical practices were followed during experimentation. The seeds of chickpea (cv. GG-2) were sown in each plot of 1.80 x 3.00 m size keeping 30 × 10 cm spacing, 10 days after respective treatments. The irrigations were given as and when required. Insecticidal measures were also carried out as and when required. Observations were recorded on root rot incidence at flowering stage to harvest. Root rot incidence (%) in each treatment was calculated using following formula.

$$\text{Root rot Incidence (\%)} = \frac{\text{Number of diseased plant}}{\text{Total Number of observed plants}} \times 100$$

Experimental Details:

Design	:	RBD
Treatment	:	10
Replication	:	3
Year	:	2010-2011
Location	:	Agronomy Farm B. A. College of Agriculture Anand Agricultural University Anand – 388 110
Season	:	<i>Rabi</i>
Crop	:	Chickpea (<i>Cicer arietinum</i> L.)
Variety	:	GG - 2

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Seed rate	:	60 kg/ha
Spacing	:	30 × 10 cm
Gross plot size	:	1.80 × 3.00 m
Net plot size	:	1.20 × 2.80 m
Fertilizers	:	

	N	P	K	kg/ha
Treatment details:	20	40	-	

Tr. No.	Treatments
T ₁	Seed treatment with <i>T. viride</i> @ 5 gm/kg seed at the time of sowing
T ₂	Seed treatment with <i>T. harzianum</i> @ 5 gm/kg seed at the time of sowing
T ₃	Soil application of <i>T. viride</i> @ 5 kg/ha in 500 kg FYM/ha in the furrow 10 days prior to sowing
T ₄	Soil application of <i>T. viride</i> @ 5 kg/ha in 500 kg castor cake/ha in furrow 10 days prior to sowing
T ₅	Soil application of <i>T. harzianum</i> @ 5 kg/ha in 500 kg FYM/ha in furrow 10 days prior to sowing
T ₆	Soil application of <i>T. harzianum</i> @ 5 kg/ha in 500 kg castor cake/ha in furrow 10 days prior to sowing
T ₇	Seed treatment of Carbendazim @ 3 gm/kg seed at the time of sowing
T ₈	Seed treatment with <i>T. viride</i> @ 8 gm/kg seed with soil application of <i>Pseudomonas fluorescens</i> @ 2.5 kg/ha in furrow at the time of sowing
T ₉	Seed treatment with Carbendazim (12%) + mancozeb (63%) @ 3 gm/kg seed at the time of sowing
T ₁₀	Control (Untreated)

Note:

Treatments T₃ to T₆ were applied in split doses.

1. At the time of sowing

2. At 30 DAS without FYM and castor cake.

Observations recorded:

Observations were recorded on

1. Disease Incidence at flowering to pod filling stage

2. Gram yield (Kg/ha)

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Root rot incidence (%) in each treatment was calculated by using following formula.

$$\text{Root rot Incidence (\%)} = \frac{\text{Number of diseased plant}}{\text{Total Number of observed plants}} \times 100$$

3.10 Statistical analysis:

Statistical analysis of the data was carried out as per the procedure described by Steel and Torrie (1980).

IV. RESULTS AND DISCUSSION

Experimental results on various aspects under the study are presented and discussed.

4.1 Pathological investigations

4.1.1 Isolation, pathogenicity and confirmation of the pathogen

Chickpea plants showing the typical root rot symptoms were collected from the fields and subjected to isolate by tissue isolation technique. Initially the isolated fungus revealed dirty white growth which turned to fluffy black on 4th day of incubation on PDA medium (Plate 1). Later on at 5th day of incubation minute black sclerotial bodies formed on PDA plates. The culture was further purified by single hyphal tip method and maintained on PDA slants for further investigations.

4.1.2 Pathogenicity:

The pathogenicity of the isolate was carried out by sterilized soil inoculation method. The results observed during the study revealed that the disease was produced successfully in the sterilized inoculated soil. The typical visible symptoms (Plate 2) of root rot showing gradual yellowing and straw colour of leaves from lower to upper leaves after 50-60 days of soil inoculation. The root portion became black in colour. The

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collar region showed brownish in colour coupled with shredding of outer sheath of roots. Such plants could be easily pulled out. Uninoculated soil (control) did not show the root rot symptoms (Plate 2).

The reisolation from the artificially inoculated chickpea plants yielded pure culture of *Macrophomina* sp. identical to original in all the respect. Thus, the pathogenicity of the isolate was confirmed by soil inoculation method employing Koch's postulate.

Goyal and Jain (1998) reported that in sterilized soil inoculation, highest pre (48.10%) and post emergence (37.33%) mortality was observed with *M. phaseolina* and *Fusarium* sp., respectively as compared to seed inoculation, while Singh and Sirohi (2003) proved the pathogenicity of chickpea isolate *M. phaseolina* by using unsterilized soil in pot conditions.

4.1.3 Identification of the pathogen:

The morphological and cultural characters of the isolated *Macrophomina* sp. was studied and described as under.

4.1.3. A Mycelium:

The colonies grew very fast on PDA and attained a diameter of 90 mm in 5 days at $28\pm 2^{\circ}\text{C}$ temperature. The mycelial growth was linear, fluffy and initially dirty white in colour when young, but later turned blackish in colour producing black microsclerotia. The hyphal

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branch was at the right angles with a constriction of hyphal branches at their points of origin while, the mycelial septa was placed closed to the point of branching (Plate 3).

4.1.3. B Sclerotia:

The microsclerotia were black and variable in the size of 90-140 μm (Plate 3), where as the pycnidia were not observed on the medium.

The above cultural and morphological characters of *Macrophomina* sp. isolated from the root rot infected chickpea plants in this study were closely identical with description of *M. phaseolina* (Tassi) Goid. given by Shekhar *et al.* (2006) and Mbaye (2007).

Further the pure culture was sent for identification to Indian Type Culture Collection, Division of Plant Pathology, IARI, New Delhi-110 012 and was identified as *Macrophomina phaseolina* (ITCC No. 7810.10). Thus, the isolate *Macrophomina* sp. under study was identified and confirmed as *Macrophomina phaseolina* (Tassi) Goidanch.

4.2 Studies on variability among the chickpea isolates of *M. phaseolina*

The pathogen was isolated from all the infected chickpea root samples collected from 4 places of middle Gujarat viz. Arnej, Tarapur, Dahod and Anand. The results summarized as under.

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4.2. A Mycelial Growth:

The cultural characteristics of various 4 isolates indicated variability in fungal colony (Table 4.1). The isolates were significantly differed in mycelial growth formation (Fig. 1) and revealed that whitish dull mycelial growth was higher in isolate Mp1 (14.00 mm) on the 1st day of incubation whitish dull mycelial growth was higher in isolate Mp1 followed by Mp4 (11.50 mm). On 5th day of incubation, mycelial growth was higher in Mp4 (71.25 mm) followed by Mp1 (70.75 mm). However, fully black growth with dark black microsclerotia observed on 7th day of incubation (90.00 mm) in all 4 isolates of *M. phaseolina*.

Table 4.1: Mycelial growth of *M. phaseolina* isolates on 1st, 5th and 7th day of incubation

Sr. No.	Isolates	Growth diameter (mm)*			Growth diameter mean (mm)
		1 st day	5 th day	7 th day	
1	Mp1	14.00	70.75	90.00	58.25
2	Mp2	6.25	63.75	90.00	53.33
3	Mp3	6.75	67.50	90.00	54.75
4	Mp4	11.50	71.25	90.00	57.58
Mean (mm)		9.63	68.31	90.00	
		F	C	F × C	
	SEM±	0.285	0.376	0.494	
	CD at 5%	0.826	1.301	1.431	
	CV (%)	1.76			

*Average of four replications.

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4.2. B Sclerotial formation:

The study on variability among the 4 isolates (Table 4.2 and Fig. 2) revealed that there was no sclerotial formation on 1st day of incubation in none of the isolates. As on 5th day of incubation, the number of sclerotia were formed minimum in isolate Mp 1 (52 No.) and maximum in the isolate Mp 4 (62 No.). However, on the 7th day of incubation, the maximum number of sclerotia observed in the isolate Mp1 (75.25 No.) and lowest in isolate Mp 2 (68.50 No.).

Table 4.2: Sclerotial formation of different isolates on 1st, 5th, and 7th day of incubation

Sr. No.	Isolate	No. of sclerotial formation*			No. of sclerotial formation (mean)
		1 st day	5 th day	7 th day	
1	Mp1	0.00	52.0	75.25	42.42
2	Mp2	0.00	52.0	68.50	40.17
3	Mp3	0.00	56.0	70.25	42.08
4	Mp4	0.00	62.0	71.75	44.58
No. of sclerotial formation mean		0.00	55.50	71.44	
		F	C	F × C	
	SEM±	0.592	0.353	1.025	
	CD at 5%	1.716	1.223	2.973	
	CV (%)	4.85			

*Average of four replications.

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4.2. C Temperature:

To determine the optimum temperature for the mycelial growth of different 4 isolates of *M. phaseolina*, it was grown at different 5 temperature levels on PDA medium. The results presented (Table 4.3 and Fig. 3) indicated that the mycelial growth of each isolate of *M. phaseolina* was significantly grew best at 30° C temperature (88.75 to 89.7 mm) followed by 35° C temperature (84.75 to 87.25 mm) on 7th day of incubation. There was very poor mycelial growth (22.50 to 30.75 mm) at 15° C temperature in all the test isolates on 7th day of incubation.

Table 4.3: Mycelial growth of *M. phaseolina* isolates at different temperature *in vitro*

Sr. No	Isolates	Temperature/ Mycelial growth (mm)*				
		15° C	20° C	25° C	30° C	35° C
1	Mp1	25.75	55.25	71.25	89.75	84.75
2	Mp2	22.50	55.25	69.75	88.75	86.50
3	Mp3	29.75	50.50	75.75	89.50	87.00
4	Mp4	30.75	52.00	75.50	89.25	87.25
SEM±		0.924	1.229	1.111	0.525	0.963
CD at 5%		2.845	3.787	3.424	-	-
CV (%)		6.81	4.62	3.04	1.18	2.23

*Average of 4 replications.

Jha and Sharma (2005) reported that the optimum temperature for *R. bataticola* was 30-35°C, both for mycelial growth and sclerotial formation. At 40°C temperature, very poor mycelial growth and

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no sclerotium was observed in their study. Sandhu and Singh (1999) and Sharma *et al.* (2004) also reported higher temperature ranges of 25-30°C favored the growth of *M. phaseolina*. Naik *et al.* (2010) also reported that 30°C and 35°C was congenial for optimum mycelial growth (90 mm and 89.50 mm, respectively) of *M. phaseolina*.

4.3 Effect of solid culture media on the growth of *M. phaseolina in vitro*

Six different solid media were tested for their suitability for the mycelial growth and sclerotial production of the test pathogen. The data are presented in Table 4.4, Fig. 4 and Plate 4.

Among the 6 solid media tested, significantly the highest mycelial growth was recorded on PDA (88.25 mm) followed by Richard's agar (83.63 mm). The next best media in order of merit were Oat Meal Agar (73.75 mm), Czapek's agar (62.50 mm) and Peptone sucrose agar (59.25 mm) medium. Significantly the lowest growth (52.50 mm) was recorded in Sabouraud agar medium.

The excellent sclerotial formation of *M. phaseolina* was observed on PDA and Richard's agar medium. Good sclerotial formation recorded on Oat meal agar, while moderate sclerotial formation was

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observed on Czapek's agar and Sabouraud agar. Peptone Sucrose Agar revealed very poor sclerotial formation.

The results of present investigation are similar to the results obtained by earlier workers. Grover and Sakhuja (1981) reported that pycnidial stage was never produced in artificial media. Sobti and Sharma (1992) reported that the fungus well grown on PDA medium with similar cultural characteristics of mycelium and sclerotial formation were found after 3rd and 7th day of incubation, respectively confirmed our results. Suriachandraselvan and Seetharaman (2003) also reported the best radial growth and excellent sclerotial formation of *M. phaseolina* on PDA and Richard's agar medium.

Table 4.4: Mycelial growth and sclerotial formation of *M. phaseolina* on different solid media *in vitro*

Sr. No.	Medium	Average colony diameter (mm)*	Sclerotial formation**
1	Czapek's Agar	62.50 ^d	++
2	Oat Meal Agar	73.75 ^c	+++
3	Peptone Sucrose Agar	59.25 ^d	+
4	Potato Dextrose Agar	88.25 ^a	++++
5	Richard's Agar	83.63 ^b	++++
6	Sabouraud's Agar	52.50 ^e	++
	SEM±	1.37	
	CD at 5%	4.06	
	CV (%)	3.91	

Note: Treatment means with the letter in common are not significant by DNMR test at 5% level of significance

* Average of 4 replications

** + Poor, ++ Moderate, +++ Good, ++++ Excellent

4.4 Studies on antagonism between root rot pathogen and antagonists *in vitro*:

The hazardous effects of chemicals used in plant disease management have diverted Plant Pathologists to find out an effective alternative method with little or no adverse effect on environment. Notable success of disease control through the use of antagonistic microorganisms in the laboratory, glass house and field have been achieved during past several years and based on this information, there is a possibility of developing biological control for plant diseases management. Now a day the commercial formulations of some of the bio-control agents have already been available in the market. However, inadequate information on the antagonists under varying conditions is a major constraint in the large scale adoption of this technology.

In the study of screening of antagonists against *M. phaseolina* 5 fungal and 2 bacterial antagonists were screened by dual culture technique *in vitro*.

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The results presented in Table 4.5, Fig. 5 and Plate 5 revealed that all the antagonists screened against *M. phaseolina* were significantly superior showing growth inhibition of 25.83 to 35.45 % over the control. Out of 7 antagonists, *T. harzianum* showed significantly maximum per cent growth inhibition (35.45%) with lowest fungal colony diameter (58.25 mm) of *M. phaseolina* followed by *T. viride* (32.36%) with the fungal colony diameter (60.88 mm). The next best antagonists in order of merit were *T. virens* with 31.11% and *A. niger* 30.41% growth inhibition, respectively with (62.00 mm) and (62.63 mm) fungal colony diameter. *A. flavus* exhibited moderate inhibition (27.77 %) with fungal colony diameter (65.00 mm) of *M. phaseolina*.

In case of bacterial antagonists, *P. fluorescens* showed significantly maximum per cent growth inhibition (29.02%) followed by *B. subtilis* with the lowest per cent growth inhibition (25.83%) of *M. phaseolina*.

Among the antagonists evaluated, all the 3 species of *Tichoderma* proved highly antagonistic against *M. phaseolina*.

Table 4.5: Antagonism of different bioagents against *M. phaseolina* under dual culture technique *in vitro*

Sr. No.	Test organisms	Average colony diameter of pathogen (mm)*	Growth Inhibition (%)
1	<i>Aspergillus flavus</i>	65.00 ^{ab}	27.77

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2	<i>Aspergillus niger</i>	62.63 ^{bcd}	30.41
3	<i>Bacillus subtilis</i>	66.75 ^a	25.83
4	<i>Pseudomonas fluorescens</i>	63.88 ^{bc}	29.02
5	<i>Trichoderma harzianum</i>	58.25 ^c	35.45
6	<i>Trichoderma virens</i>	62.00 ^{cd}	31.11
7	<i>Trichoderma viride</i>	60.88 ^d	32.36
	SEM±	0.74	-
	CD at 5%	2.18	-
	CV (%)	2.26	-

Note: Treatment means with the letters in common are not significant by DNMRT test at 5% level of significance

*Average of four replications

Present findings are in conformity with earlier research workers. Rajeshwari *et al.* (1998) also observed wide inhibition zone of *M. phaseolina* by *T. harzianum* and *T. viride*. Ahamad and Shrivastava (2000) reported inhibition of growth and sclerotial production of *M. phaseolina* by *T. viride* and *B. subtilis*. Desai and Kulkarni (2002) found *P. fluorescens* effective against *M. phaseolina*. Inhibition of mycelial growth and sclerotial formation of *R. bataticola* of chickpea was reported by Sindhan *et al.* (2002) and Kasawate *et al.* (2003). Suriachandraselvan *et al.* (2004) reported that *T. viride* and *T. harzianum* significantly inhibited the mycelial growth of *M. phaseolina* (73-74%). Naik *et al.* (2010) reported *T. harzianum* as the most effective antagonistic agent, while *T. viride* and *T. virens* as second best against the pathogen causing chickpea root rot disease.

4.5 Bio-efficacy of fungicides against *M. phaseolina*

in vitro

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Nine fungicides were tested against the pathogen by “poisoned food technique” at three different concentrations under *in vitro* condition. The data presented in Table 4.6; Fig. 6 (a), (b) and (c) and Plate 6 (I), (II) and (III) revealed that the significant reduction in radial growth diameter was found at all the concentrations of various fungicides. Among the fungicides, bavistin, sixer, cosko, tilt, amistar Top, thirum and vitavex at all the 3 concentrations completely inhibited the mycelial growth of *M. phaseolina*. The next best fungicides in order of efficacy was topsin M @ 750 ppm (82.22 % growth inhibition) followed by topsin M @ 500 and 250 ppm, respectively. These two concentrations were at par with each other and recorded 80.83 and 77.22 per cent growth inhibition, respectively. Amistar Top @ 750 ppm recorded 28.05 % growth inhibition and found at par with amistar @ 500 and 250 ppm with 25.83 and 21.11 per cent growth inhibition, respectively.

Present findings are in agreement with Mujumdar and Som (1987), Lambhate *et al.* (2002) and Jha and Sharma (2006a). They reported that among the fungicides tested *in vitro*, carbendazim (Bavistin) @ 0.05% as well as @ 0.1, 0.2 and 0.3% concentrations completely inhibited (100%) mycelial growth of *M. phaseolina*.

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Prajapati *et al.* (2002) and Chattopadhyay and Shashtry (2002) reported that thirum @ 100 ppm completely inhibited *M. phaseolina*. Prajapati *et al.* (2002), Chattopadhyay and Shashtry (2002) and Surywanshi *et al.* (2008) reported carbendazim alone, carbendazim alone with thirum and carboxin @ 0.02, 0.03 and 0.05% concentrations, respectively completely inhibited the radial growth of *M. phaseolina in vitro*.

Prajapati *et al.* (2002) and Lambhate *et al.* (2002) reported that topsin M @ 0.1, 0.2 and 0.3% concentrations completely inhibited the mycelial growth of *M. phaseolina*. Sudhakar *et al.* (2004) reported cent percent inhibition of *M. phaseolina* with carboxin @ 500 and 1000 ppm concentrations.

Table 4.6: Effect of fungicides on per cent growth inhibition of *M. phaseolina* at different concentrations *in vitro*

Sr. No.	Treatment	Concentration (ppm)	Radial growth (mm)*	Per cent growth inhibition
1	Bavistin (50%WP)	250	0	100
		500	0	100
		750	0	100
2	Sixer (75% WP)	500	0	100
		750	0	100
		1000	0	100
3	Cosko (75%SD)	300	0	100
		500	0	100
		700	0	100
4	Tilt (25% EC)	25	0	100
		50	0	100
		75	0	100
5	Amistar Top (0.125 %)	50	0	100
		100	0	100
		150	0	100

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6	Thiuram (75%WP)	500	0	100
		750	0	100
		1000	0	100
7	Topsin M (70%WP)	250	20.50	77.22
		500	17.25	80.83
		750	16.00	82.22
8	Vitavax (75% WP)	250	0	100
		500	0	100
		750	0	100
9	Amistar (23 SC)	250	71.00	21.11
		500	66.75	25.83
		750	64.75	28.05
10	Control	-	90	-
SEM±		-	0.227	-
CD at 5%		-	0.634	-
CV (%)		-	3.66	-

*Average of four replications.

4.6 Studies on effect of herbicides on the growth of *M. phaseolina in vitro*:

Four different herbicides were screened *in vitro* for their comparative efficacy against *M. phaseolina* by poison food technique.

4.6. A Herbicides:

The results presented in Table 4.7, Fig. 7 and Plate 7 (I), (II) and (III) revealed that all the 4 herbicides showed significant inhibitory effect against mycelial growth as compared to control. The maximum inhibition was observed in alachlor (100%) and in pendimethalin (98.42%) at all the three concentrations. Significantly lower growth inhibition was observed in atrazin (33.31%) and in 2,4 -D (15.92%) than rest of the herbicides.

4.6. B Concentrations:

The growth inhibition was increased significantly with the increase in concentrations of the herbicides. In alachlor all the three

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concentrations gave 100 per cent growth inhibition. At 100 ppm concentration, the per cent growth inhibition was found less as compared to 200 ppm concentration showing maximum inhibition in the range of 26.38 -100 per cent. However, at 150 ppm concentration, the per cent growth inhibition was moderate i.e. 9.44 -100 per cent, respectively in each herbicide.

4.6. C Interaction:

The interaction effect of the herbicides v/s concentrations revealed that at lower (100 ppm) concentration, pendimethalin showed significantly highest (95.27%) per cent growth inhibition of *M. phaseolina*, while at medium concentration (150 ppm) pendimethalin showed 100% growth inhibition as compared to the rest of the herbicides *in vitro*.

Present findings are in agreement with Lima *et al.* (1997). They reported that alachlor @ 1000 and 2000 ppm inhibited the mycelial growth and sclerotial formation of *M. phaseolina*. Jha and Sharma (2006b) also reported that alachlor at 100 and 200 ppm inhibited the mycelial growth of *M. phaseolina*.

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Table 4.7: Effect of herbicides on mycelial growth of *M. phaseolina* in vitro

Sr. No.	Treatments	Growth inhibition (%)*			Growth inhibition mean (%)
		Concentration (ppm)			
		100	150	200	
1	Alachlor 50 EC	100	100	100	100
2	Atrazin 50% WP	24.94	31.38	43.61	33.31
3	Metribuzin 70% WP	2.50	31.94	26.38	20.27
4	Pendimethalin 30EC	95.27	100	100	98.42
5	2,4 - D 80% WP	5.83	9.44	32.50	15.92
		F	C	F × C	
	SEM±	0.359	0.279	0.622	
	CD at 5%	1.022	0.791	1.769	
	CV (%)	2.96			

*Average of four replications.

4.7 Studies on influence of sowing dates, soil moisture levels and incidence of chickpea root rot in pot conditions

Four different sowing dates with 4 different soil moisture levels were tested against root rot disease incidence in pot conditions.

The results presented in Table 4.8 and Fig. 8 revealed that all 4 sowing dates and different soil moisture levels (25, 50, 75 and 100%) showed significant effect on root rot incidence. The mean disease

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incidence was found 72.68 % and 67.50 % at water stress conditions i.e 25% and 50% soil moisture, where as the incidence was least (1.28%) at 75% and 100% soil moisture levels. However, the interaction effect was found un individual to different dates of sowing and soil moisture levels.

Table 4.8: Effect of different sowing dates and different soil moisture levels on root rot incidence in pot culture

Sr. No.	Treatments (Sowing dates)	Moisture level (%)*				Root rot incidence (%) mean
		Root rot incidence (%)				
		25	50	75	100	
1	15 th October	72.58 (91.03)	67.49 (85.34)	1.28 (0.05)	1.28 (0.05)	35.65
2	1 st November	74.79 (93.11)	68.32 (86.35)	1.28 (0.05)	1.28 (0.05)	36.42
3	15 th November	69.70 (87.96)	65.89 (83.31)	1.28 (0.05)	1.28 (0.05)	34.54
4	1 st December	73.65 (92.07)	68.32 (86.35)	1.28 (0.05)	1.28 (0.05)	36.13
Root rot incidence (%) Mean		72.68	67.50	1.28	1.28	
		F	C	F × C		
	SEM±	0.485	0.485	0.970		
	CD at 5%	1.371	1.371	NS		
	CV (%)	6.07				

*Average of five replications

Value in out side the parenthesis are arc sin transformed value

Value in parenthesis are retransformed value

Present findings are in agreement with Husain and Ghaffar (1995) reported that colonization of *M. phaseolina* on chickpea root was higher when plants were moisture stressed as compared to unstressed plant. Patel and Anahosur (2001) evaluated different 3 dates of sowing i.e

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October 1st, 15th and November 1st in pot condition with 4 available moisture per cent and reported that lower moisture (25%) gave higher disease incidence (47.80%) and higher moisture (100%) gave lower disease incidence (1.28%) of *M. phaseolina* in chickpea.

4.8 Management of Root rot of chickpea in field conditions

The effective biocontrol agents, oil cakes and chemicals which were found promising under laboratory studies were further evaluated for the management of root rot (*M. phaseolina*) disease of chickpea under field conditions. The antagonists viz, *T. viride* and *T. harzianum* as seed dresser and in combination with FYM or castor cake and other chemicals were evaluated for their efficacy against root rot pathogen in field conditions. The observations on root rot incidence prior to harvest i.e. flowering to pod filling stage as well as chickpea yield were recorded and statistically analyzed.

The data presented in Table 4.9, Fig. 9 & 10 and Plate 8 revealed that the treatment effects were found significantly superior over control in managing the root rot of chickpea. Treatment T₃ i.e. combined soil application of *T. viride* @ 5 kg/ha + 500 kg FYM/ha showed significantly lowest disease incidence (37.27%) and it was at par with treatment T₅ i.e. combined soil application of *T. harzianum* @ 5 kg/ha + 500 kg FYM/ha with 38.82% and treatment T₄ i.e. *T. viride* 5 kg/ha + 500

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kg castor cake/ha with 41.56% disease incidence as compared to control (67.08%) treatment. The next best treatments in order of merit were T₆ i.e. *T. harzianum* @ 5 kg/ha in 500 kg castor cake/ha and T₂ i.e. seed treatment with *T. harzianum* @ 5 gm/kg seed with 44.63 and 49.24% disease incidence, respectively as compared to control treatment. However, treatment T₇ i.e seed treatment with Carbendazim @ 3 gm/kg seed (ST); T₈ i.e combined application of *T. viride* @ 8 gm/kg seed + *Pseudomonas fluorescence* @ 2.5 kg/ha and T₉ i.e seed treatment with thirum @ 3gm/kg were found un effective for managing the disease.

As far as yield concerned the treatment effects were found significantly superior over control. Treatment T₃ i.e. combined application of *T. viride* @ 5 kg/ha + 500 kg FYM/ha gave significantly maximum chickpea yield (3273.6 kg/ha) and controlled maximum root rot disease (44.43%) over control treatment and ranked first as the best treatment and found at par with treatment T₅ i.e. combined application of *T. harzianum* @ 5 kg/ha in 500 kg FYM/ha by giving yield of 2827.2 kg/ha as compared to control (1190.4 kg/ha) with 42.12% disease over control and ranked as second best treatment. Treatment T₄ i.e. combined application of *T. viride* @ 5 kg/ha + 500 kg castor cake/ha stood third highest by giving yield of 2529.6 kg/ha with 38.04% disease control. The lowest yield of 1438.4 kg/ha was found in T₉ i.e. seed treatment with thirum @ 3gm/kg to control (1190.4 kg/ha) with minimum (6.67%)

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disease over control. However, treatment T₇ i.e seed treatment with Carbendazim @ 3 gm/kg seed (ST); T₈ i.e combined application of *T. viride* @ 8 gm/kg seed + *Pseudomonas fluorescens* @ 2.5 kg/ha and T₉ i.e seed treatment with thiram @ 3gm/kg gave lower yield of chickpea as compared to other treatment.

Table 4.9: Field evaluation of different bioagents, oil cakes and fungicides against root rot disease of chickpea

Sr. No.	Treatments	Root rot incidence (%)	Yield (kg/ha)	Disease over control (%)
1	<i>T. viride</i> @ 5 gm/kg seed (ST)	50.41 ^{de*} (59.38) **	1934.4 ^{de}	27.85
2	<i>T. harzianum</i> @ 5 gm/kg seed (ST)	49.24 ^e (57.37)	2132.8 ^{cde}	26.59
3	<i>T. viride</i> @ 5 kg/ha + 500 kg FYM/ha (SA)	37.27 ^h (36.67)	3273.6 ^a	44.43
4	<i>T. viride</i> @ 5 kg/ha + 500 kg castor cake/ha (SA)	41.56 ^{fg} (44.01)	2529.6 ^{bc}	38.04
5	<i>T. harzianum</i> @ 5 kg/ha + 500 kg FYM/ha (SA)	38.82 ^{gh} (39.29)	2827.2 ^b	42.12
6	<i>T. harzianum</i> @ 5 kg/ha + 500 kg castor cake/ha (SA)	44.63 ^f (49.35)	2232.0 ^{cd}	33.46
7	Carbendazim @ 3 gm/kg seed (ST)	54.36 ^{cd} (66.04)	1785.6 ^{ef}	18.96
8	<i>T. viride</i> @ 8 gm/kg seed (ST) + <i>Pseudomonas fluorescens</i> @ 2.5 kg/ha (SA)	54.76 ^c (66.70)	1736.0 ^{ef}	18.36
9	Thiram @ 3gm/kg seed (ST)	62.54 ^b (78.73)	1438.4 ^{fg}	6.67
10	Control	67.08 ^a (84.83)	1190.4 ^g	-
	SEM±	1.26	124.05	
	CD at 5%	3.76	368.60	
	CV (%)	4.37	10.19	

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Note: Treatment means with the letters in common are not significant by DNMRT test at 5% level of significance

*Figures those outside the parenthesis are arc sin transformed values

** Figures in the parenthesis are retransformed values

SA= soil application

ST= seed treatment

The results similar to the present investigations were achieved by Singh and Sindhan (1998) reported that seed and soil application of *T. viride* with FYM reduced the root rot incidence in blackgram. Ahamad and Srivastava (2000) reported that seed treatment with *T. viride* reduced the dry root rot disease of chickpea. Sethuraman *et al.* (2003) reported that seed treatment + soil application of *T. viride* showed significantly less incidence of root rot disease in blackgram. Sharma and Gupta (2004), Rathore (2006) and Jaiman and Jain (2008) reported that seed treatment with Carbendazim showed minimum incidence of dry root rot. Gaur *et al.* (2005a and 2005b) reported that soil application of *T. harzianum* @ 10 kg/ha in 200 kg FYM gave significantly lowest mortality due to root rot of cotton followed by seed treatment with *T. harzianum* @ 10 gm/kg of seed. Naik *et al.* (2010) reported soil application of *T. viride* @ 2.5 kg/ha with FYM @ 500 kg/ha most effective in reducing the disease incidence (20.45%) and increasing the seed yield (1327.25 kg/ha) and ranked first by controlling 63.13% root rot disease incidence.

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V. SUMMARY AND CONCLUSION

Chickpea (*Cicer arietinum* L.) is the most important grain legume globally grown in over 40 countries. It occupies very important position in semi-arid farming system both for human nutrition and restoring the soil fertility and keeps the soil alive and productive. Considering the seriousness of the chickpea root rot disease caused by *Macrophomina phaseolina* (Tassi) Goid. the present investigations were undertaken with bioagents, organic manures and chemicals for the management of root rot disease at Department of Plant Pathology, B. A. College of Agriculture, Anand Agricultural University, Anand during Rabi 2009–2011. The summary of the research work carried out and conclusion drawn are described.

The freshly infected chickpea plants exhibited typical root rot symptoms were collected from the area of Anand district and were subjected to isolate and revealed the association of *Macrophomina* sp. after studying the cultural and morphological characters of the fungus.

The pathogenicity test was carried by artificial-soil inoculation method and proved Koch's Postulate. The artificial inoculated plants exhibited gradual straw coloured yellowing of leaves with dark coloured shredded outer sheath of roots. The reisolated culture of the

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fungus was identified as *Macrophomina phaseolina* (Tassi) Goidanich (ITCC No. 7810-10).

The cultural and morphological characters of the fungus revealed that *M. phaseolina* initially grew fast with dirty white mycelium later on fluffy black with dark black microsclerotia formed on PDA medium. The branching of hyphae was at right angle with constriction of hyphal branches at the point of origin with closed septum to the point of branching. The microsclerotia were dark black and varied in size with 90-140 μm .

The variability among the different isolates of *M. phaseolina* was studied to determine the growth parameter and sclerotial formation. On 1st day of incubation dull whitish mycelial growth observed higher in isolate Mp 1 (14.00 mm) and Mp 4 (11.50 mm), while on 7th day of incubation the mycelial growth was maximum in all the 4 isolates. None of the isolates showed sclerotial formation on 1st day of incubation. On 5th day of incubation, isolate Mp1 and Mp2 showed moderate sclerotial formation, where as on 7th day of incubation isolate Mp 1 and Mp 2 formed maximum sclerotia i.e 75.25 and 68.50 number, respectively. All the 4 isolates of *M. phaseolina* grew best at 30⁰ C temperature showing maximum mycelial growth in the range of 88.75 mm to 89.75 mm. However, the isolate Mp 1 grew best (89.75 mm) at 30⁰ C temperature *in*

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vitro conditions. It can be concluded that higher temperature favored the root rot disease incidence.

Different 6 solid cultural media evaluated *in vitro*, Potato Dextrose Agar and Richard's Agar found favourable for maximum mycelial growth of *M. phaseolina* (88.25 mm and 83.63 mm diameter, respectively) with excellent sclerotial formation within 7 days of incubation.

Out of 7 antagonists evaluated against *M. phaseolina*, *in vitro* by dual culture technique, *T. harzianum* significantly showed maximum antagonistic effect against the pathogen by inhibiting highest mycelial growth (35.45 %). The next best antagonist *T. viride* inhibited 32.36% mycelial growth. Whereas *T. virens*, *P. fluorescens* and *B. subtilis* found least effective for inhibiting mycelial growth. It can be concluded that the fungal antagonists *T. harzianum* and *T. viride* appeared to be highly antagonistic to the root rot of chickpea.

Nine fungicides were evaluated against chickpea root rot pathogen *M. phaseolina* by poisoned food technique at different concentrations *in vitro*. All the fungicides found significantly superior for inhibiting the mycelial growth of *M. phaseolina*. Among the all, bavistin, sixer, cosko, tilt, amistar Top, thirum and vitavex at all 3 concentrations inhibited 100% mycelial growth. While topsin M at 750 ppm showed

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82.22 % growth inhibition followed by 500 ppm and 250 ppm with 80.83 % and 77.22 % growth inhibition, respectively. Amistar at 750 ppm showed 28.05 % inhibition and found at par with 500 ppm and 250 ppm with 25.83 % and 21.11 % growth inhibition, respectively. It can be concluded from the study that the higher concentration of the fungicides obviously found more effective to inhibit the growth of the pathogen than the lower concentrations.

Bio-efficacy of 5 herbicides tested *in vitro* by poisoned food technique against root rot of chickpea at 100, 150 and 200 ppm concentrations. All the herbicides at 100, 150 and 200 ppm concentrations found significantly superior for inhibiting the mycelial growth of *M. phaseolina*. Among the all herbicides, alachlor gave significantly maximum mycelial growth inhibition (100%) at all the 3 concentrations ranking first, while pendimethalin ranked second best by inhibiting 98.42% mycelial growth in all the 3 concentrations. It can be summarized that herbicides are also effective against soil borne pathogen *M. phaseolina*, *in vitro* conditions.

The experiment on 4 sowing dates with 4 different soil moisture levels in pot culture revealed higher mean incidence per cent (72.68%) in chickpea at 25% soil moisture level and lowest (1.28%) at

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higher soil moisture level i.e 75 and 100 per cent. It can be concluded that the root rot pathogen cause heavy infection in moisture stress conditions.

To manage the root rot of chickpea in field condition, combined soil application of *T. viride* @ 5 kg/ha + 500 kg FYM/ha ranked first showing least disease incidence (37.27%) with highest grain yield (3273.6 kg/ha) and controlled 44.43% root rot disease. The second best treatment i.e. combined soil application of *T. harzianum* @ 5 kg/ha + 500 kg FYM/ha gave lower disease incidence (38.82%) and higher grain yield (2827.2 kg/ha) controlling 42.12% disease. While, combined soil application of *T. viride* @ 5 kg/ha + 500 kg castor cake/ha showed 41.56% disease incidence with yield of 2529.6 kg/ha by controlling 38.04% root rot disease. It can be concluded that root rot of chickpea effectively managed by soil application of fungal bioagents *T. viride* with FYM followed by *T. harzianum* with FYM and *T. viride* with castor cake.

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* Original not seen

DECLARATION

This is to declare that the whole of the research work reported in the thesis in partial fulfilment of the requirements for the degree of **MASTER OF SCIENCE (AGRICULTURE)** in **PLANT PATHOLOGY** by the undersigned is the result of investigations done by me under direct guidance and supervision of **Dr. V. P. Gohil**, Professor, Department of Plant Pathology, B. A. College of Agriculture, Anand Agricultural University, Anand and no part of the work has been submitted for any other degree so far.

Place: Anand

(Patel S. N.)

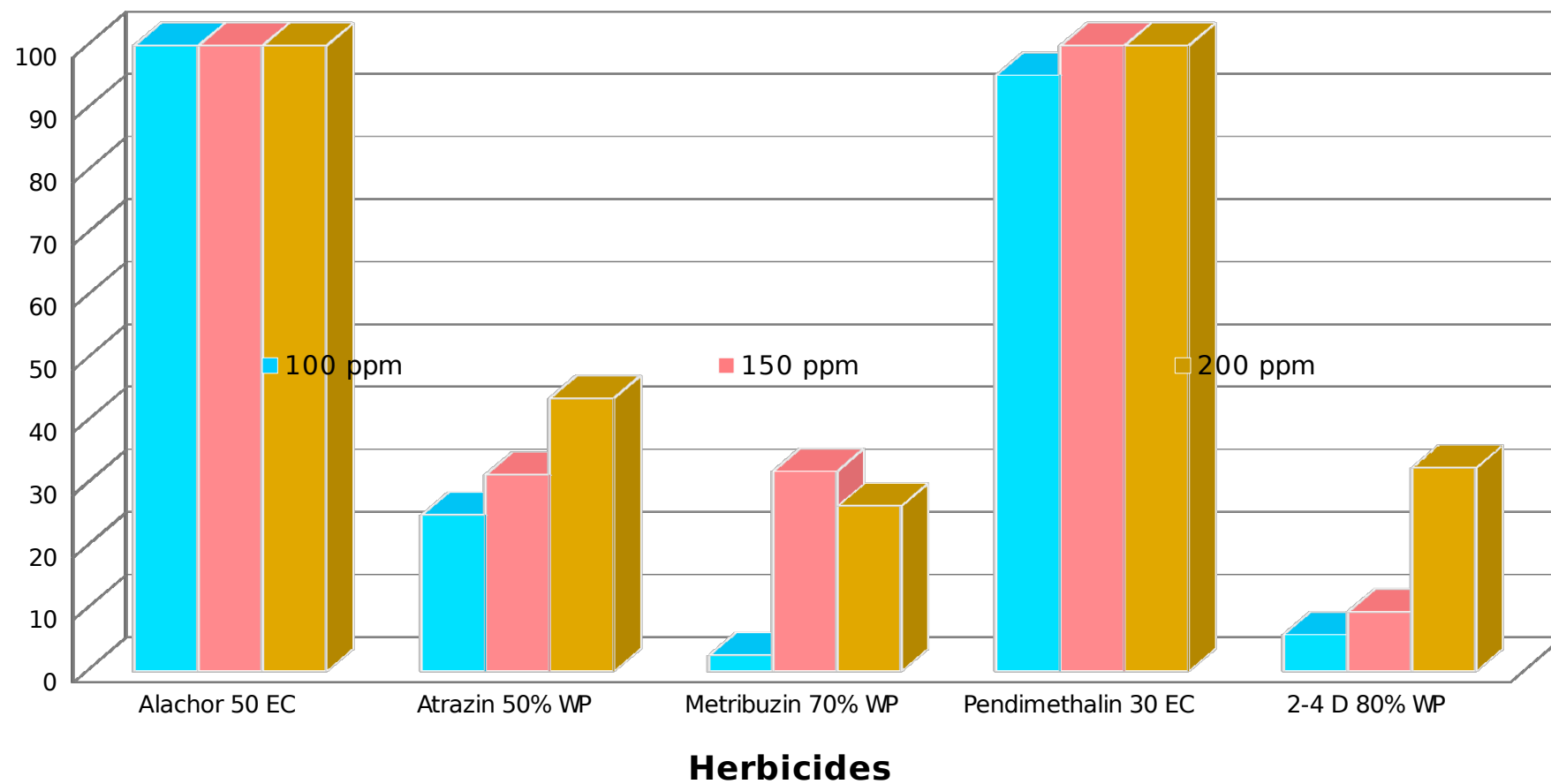
Date: / 06/2011

Countersigned by

(V. P. Gohil)

Professor
Department of Plant Pathology
B. A. College of Agriculture
A. A. U., Anand- 388 110.

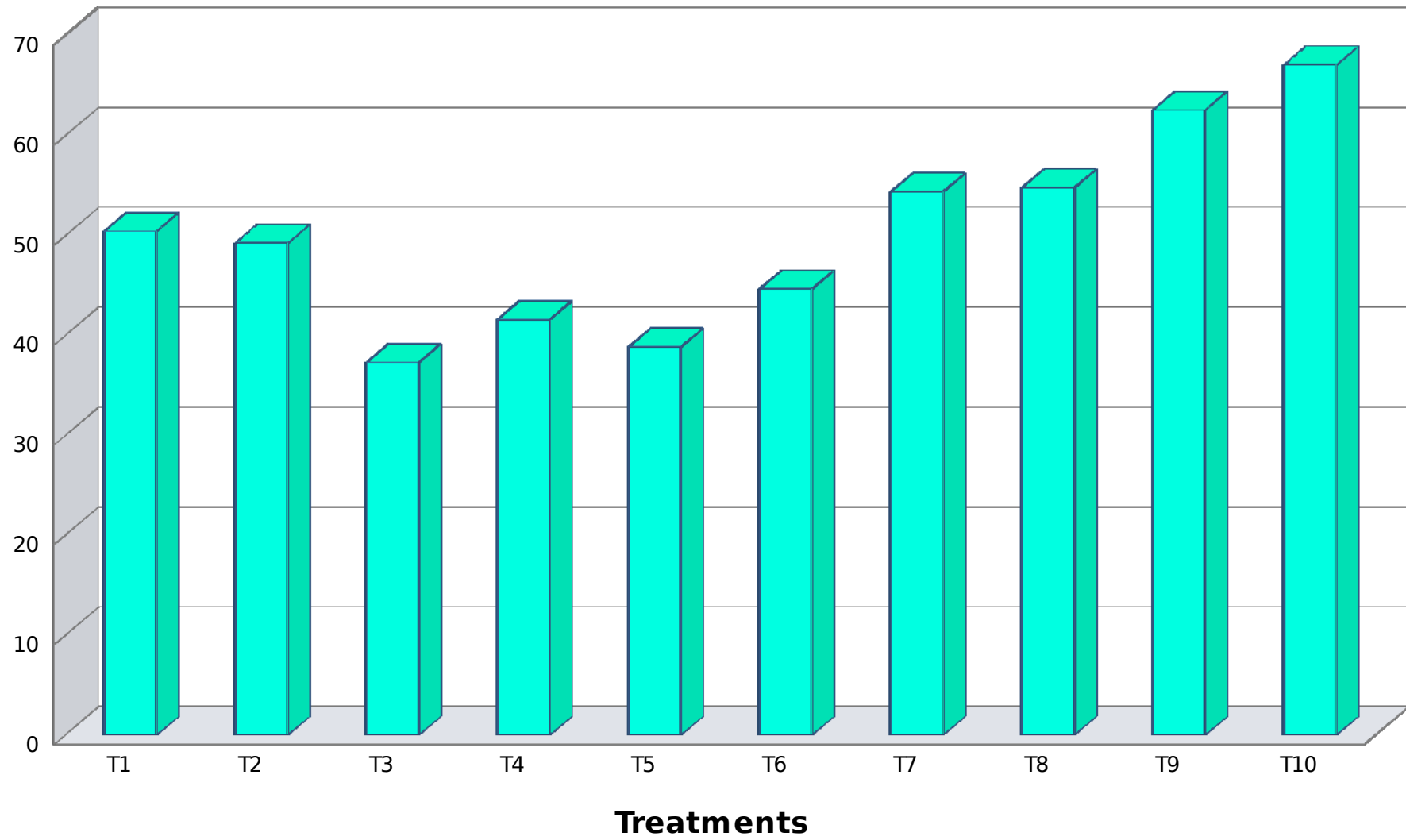
Fig. 7: Bio-efficacy of herbicides against *M. phaseolina* in vitro



%)

Alachlor 50

2,4-D 80% WP



[%]

■ 1st

■ 5th

■ 7th

1)

Isola

Fig. 1 : Mycelial growth of *M. phaseolina* isolates on 1st, 5th and

■ 1st

■ 5th

■ 7th

er)

Isolat

Fig. 2 : Sclerotial formation of *M. phaseolina* isolates on 1st, 5th

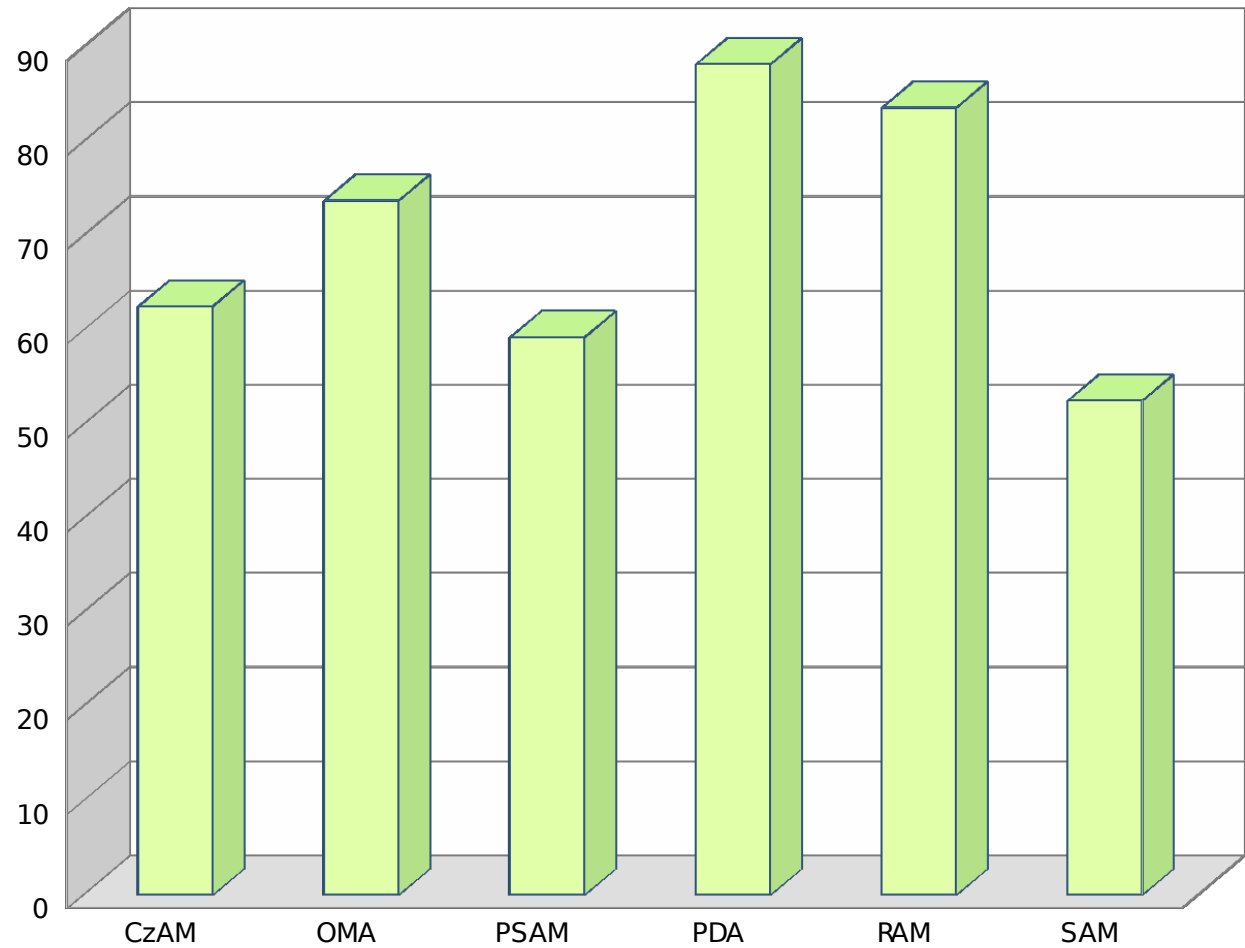
n)

■ 15o C ■ 20o C ■ 25o C ■ 30o C ■ 35o C

Isolat

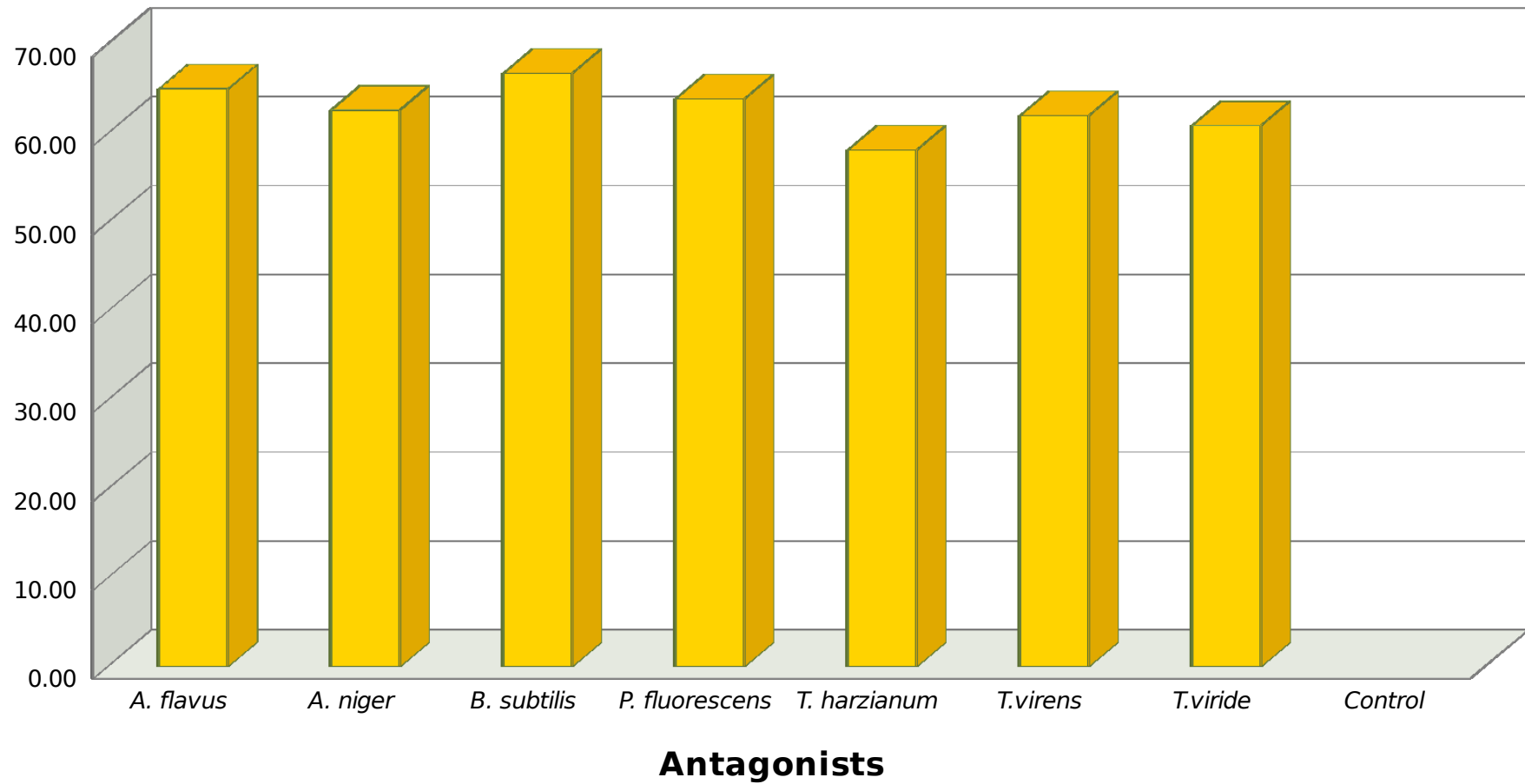
Fig. 3 : Mycelial growth of *M. phaseolina* isolates at different

Mycelial growth (mm)

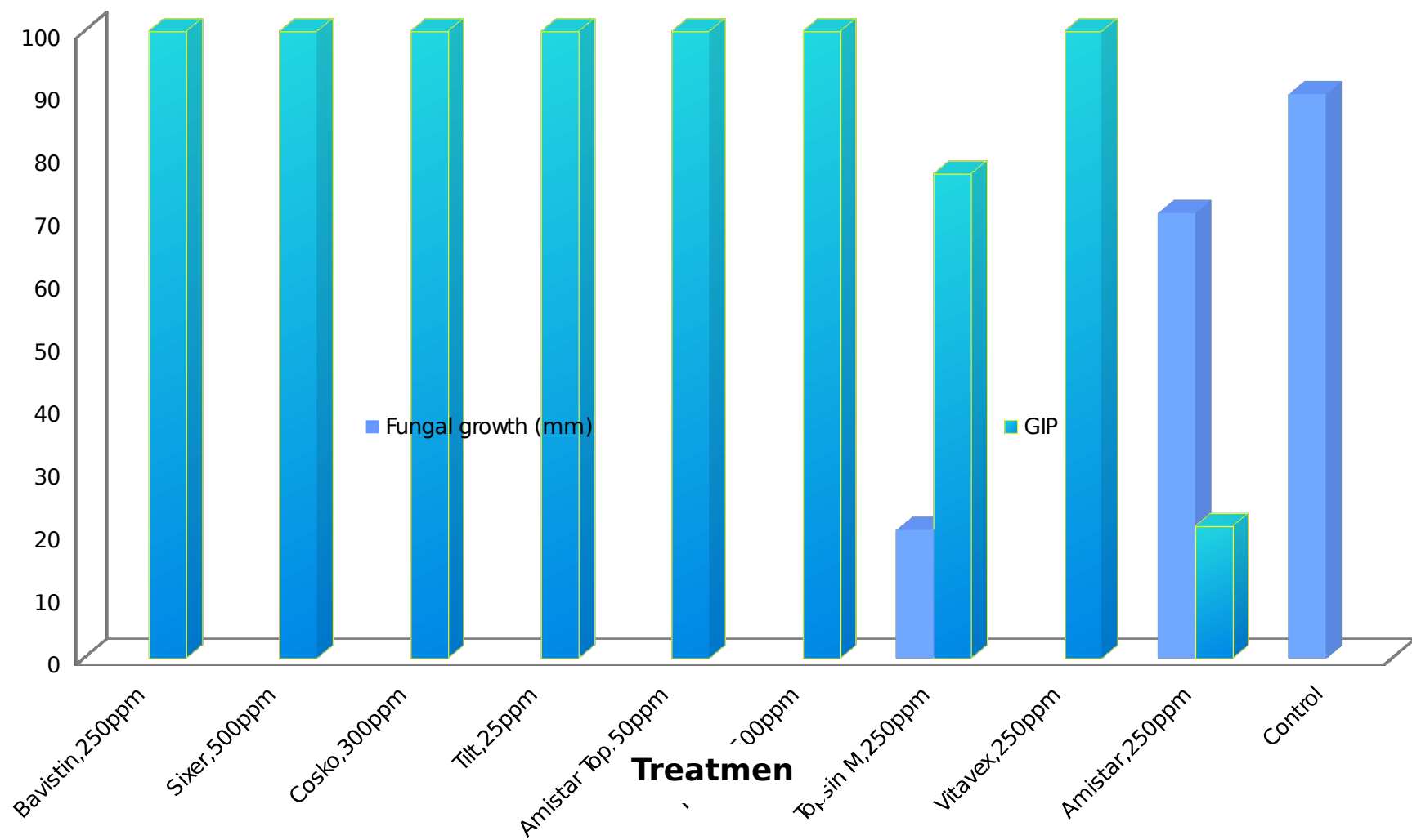


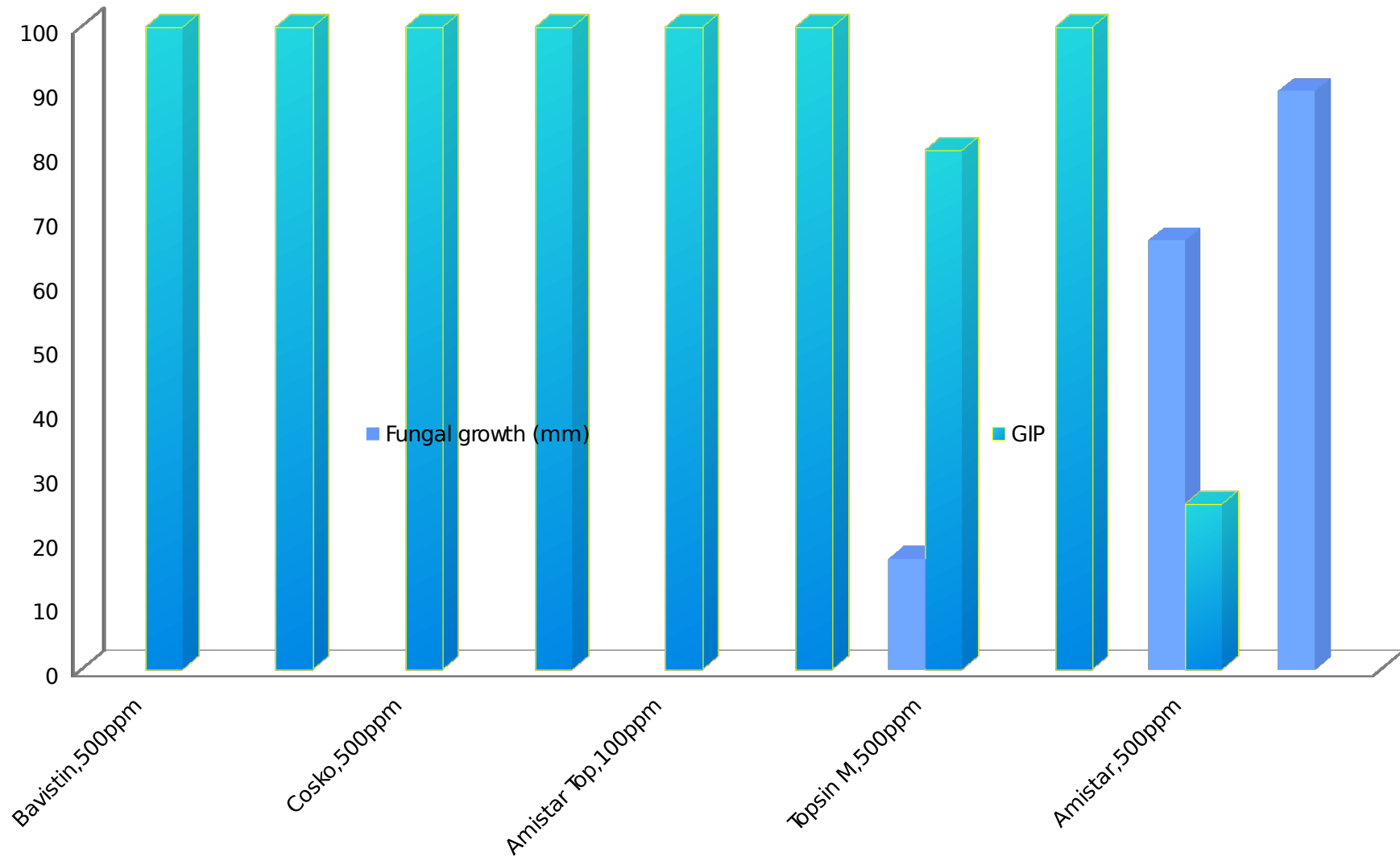
Cultural Media

Fig. 5: Studies on antagonism between *M. phaseolina* and antagonists in vitro

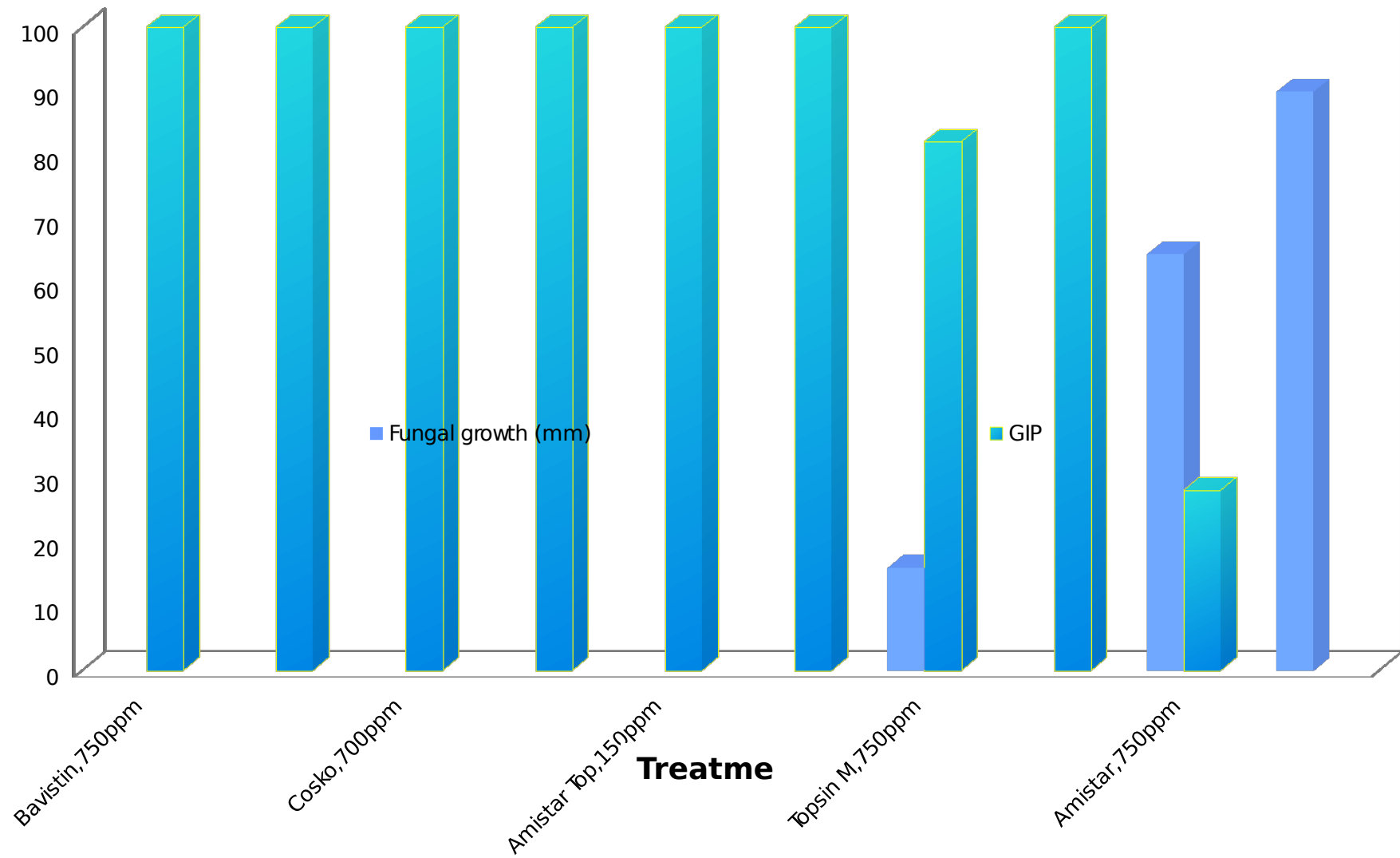


(%)





Treatmen



(%)

— Date 1

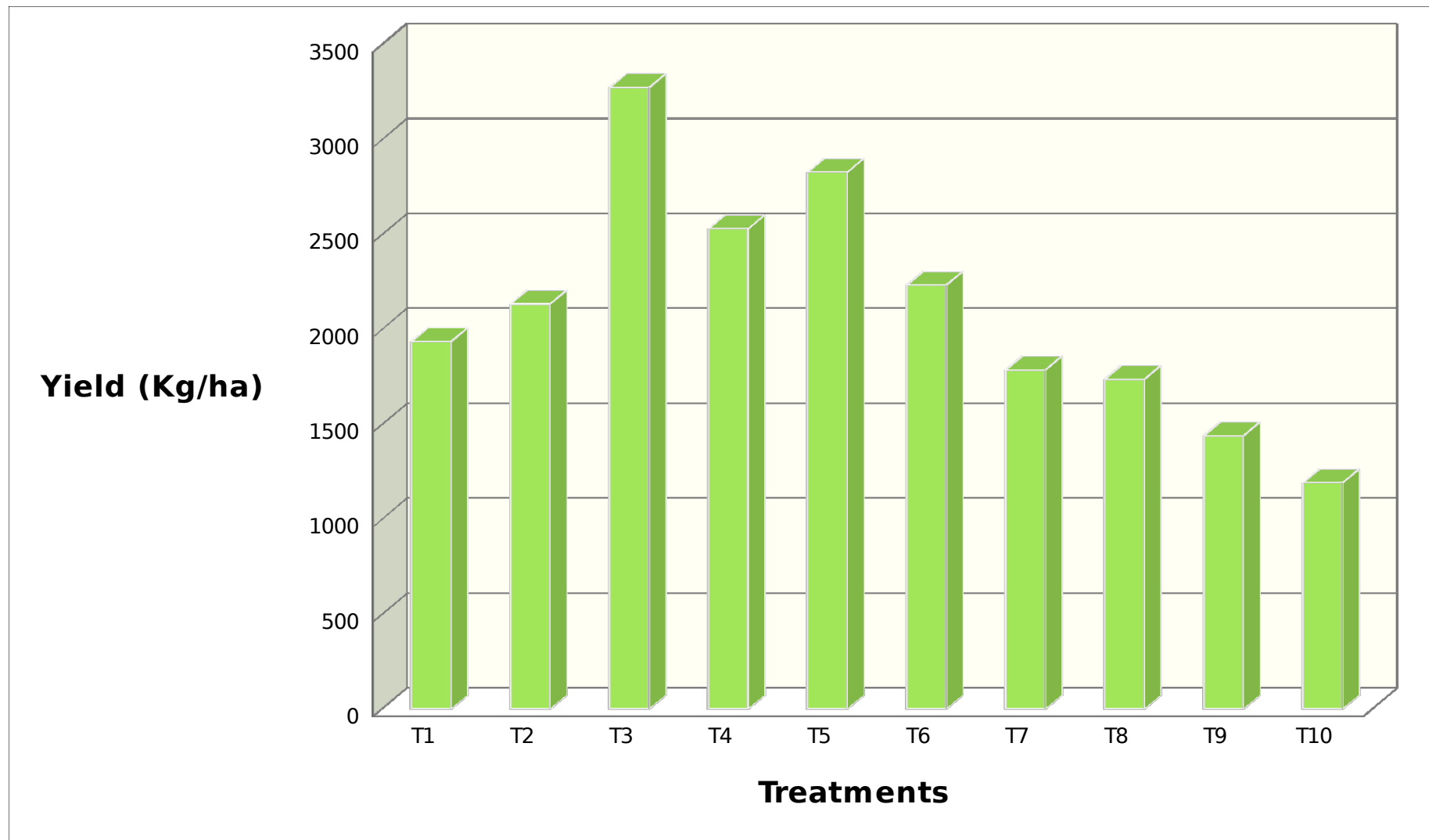
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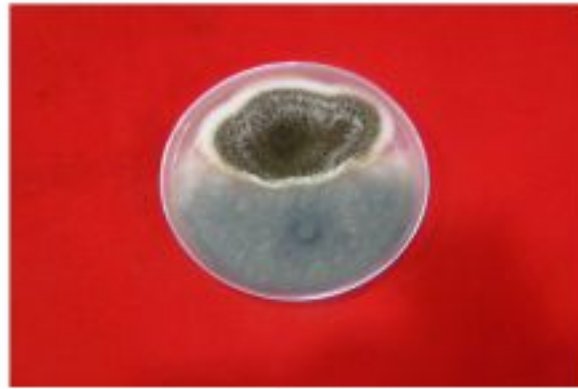
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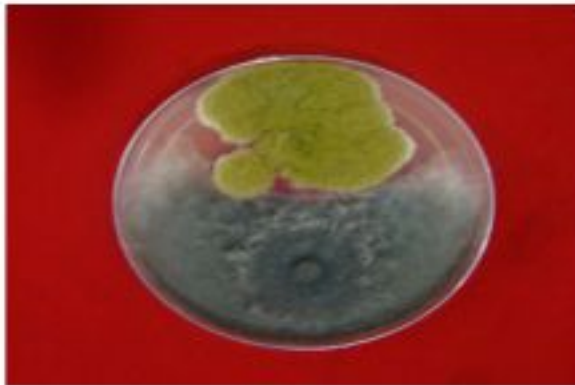
Moisture per

Fig. 8 : Effect of different sowing dates and soil moisture levels on

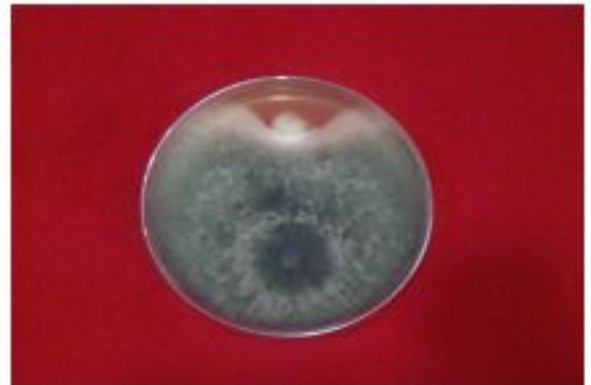




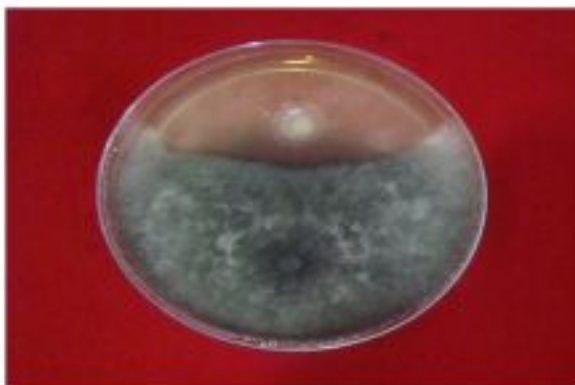
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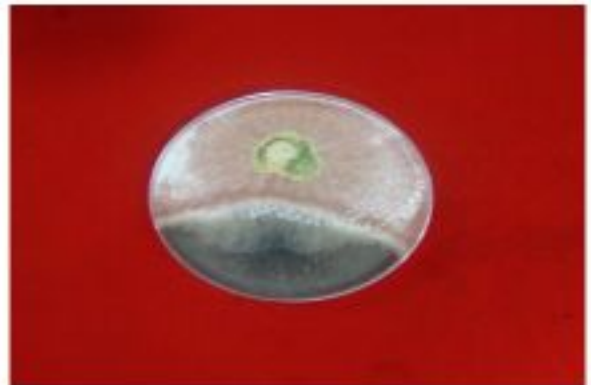
A.niger



B. subtilis



P. fluorescens



T. harzianum

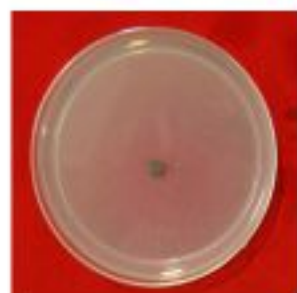


T. virens



T. viride

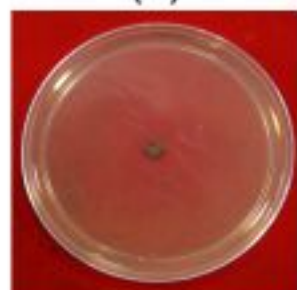
Plate 5 : Studies on antagonism between *M. phaseolina* and antagonists *in vitro*



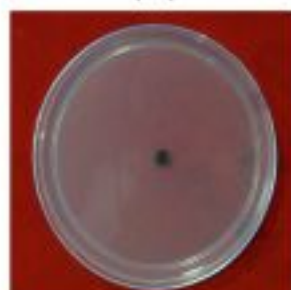
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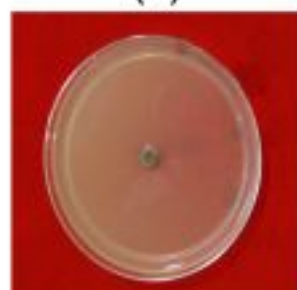
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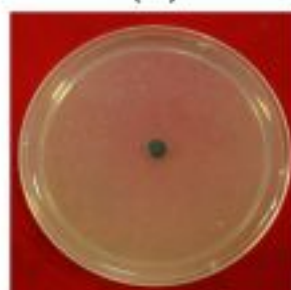
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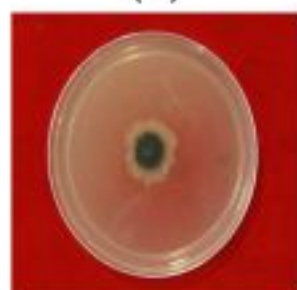
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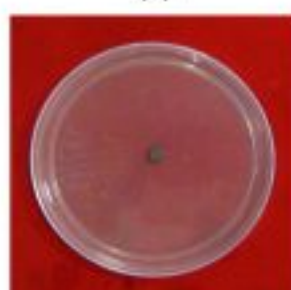
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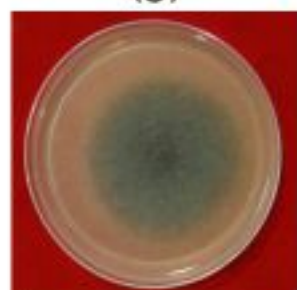
(f)



(g)



(h)



(i)



(j)

Plate 6 (I): Bio-efficacy of fungicides against *M. phaseolina* in vitro

a) Carbendazim (50% WP) @ 250 ppm

b) Sixer (75% WP) @ 500 ppm

c) Cosko (75% SD) @ 300 ppm

d) Tilt (25% SC) @ 25 ppm

e) Amistar Top (0.125%) @ 50 ppm

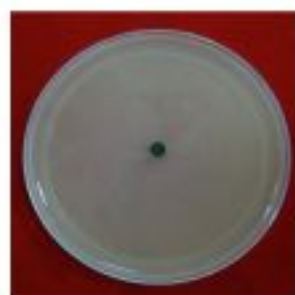
f) Thirum (75% WP) @ 500 ppm

g) Thiophanate methyl (70% WP) @ 250 ppm

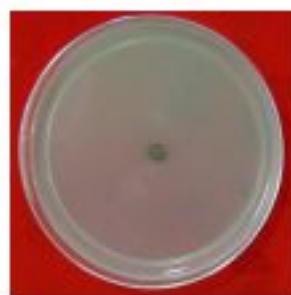
h) Vitavax (75% WP) @ 250 ppm

i) Amistar (0.1%) @ 250 ppm

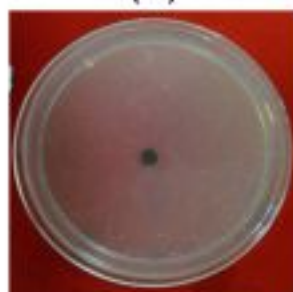
j) Control



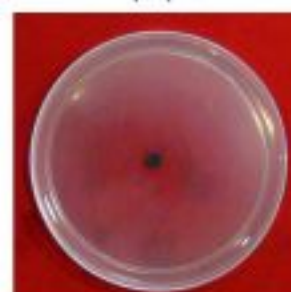
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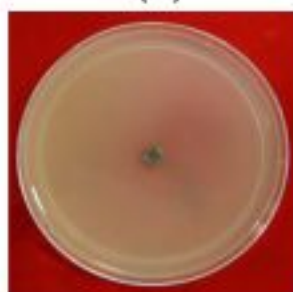
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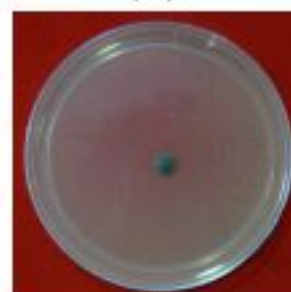
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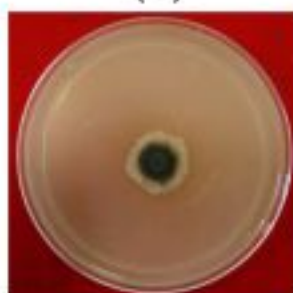
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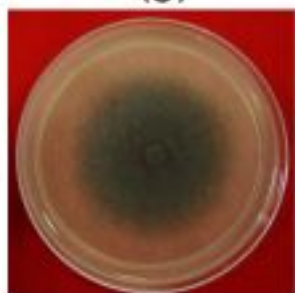
(f)



(g)



(h)



(i)



(j)

Plate 6 (II): Bio-efficacy of fungicides against *M. phaseolina* in vitro

a) Carbendazim (50% WP) @ 500 ppm

b) Sixer (75% WP) @ 750 ppm

c) Cosko (75% SD) @ 500 ppm

d) Tilt (25% SC) @ 50 ppm

e) Amistar Top (0.125%) @ 100 ppm

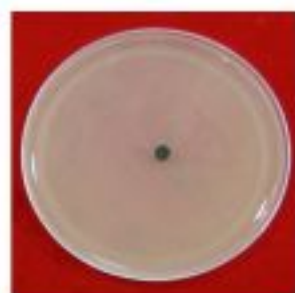
f) Thirum (75% WP) @ 750 ppm

g) Thiophanate methyl (70% WP) @ 500 ppm

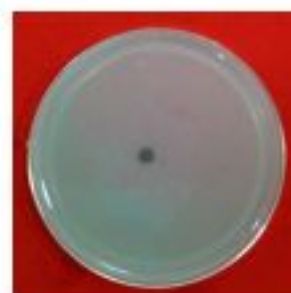
h) Vitavax (75% WP) @ 500 ppm

i) Amistar (0.1%) @ 500 ppm

j) Control



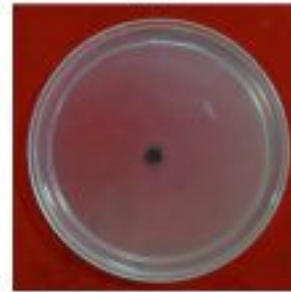
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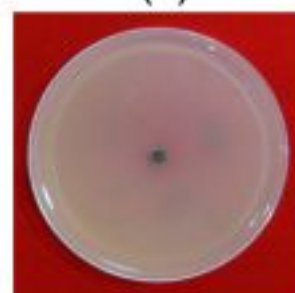
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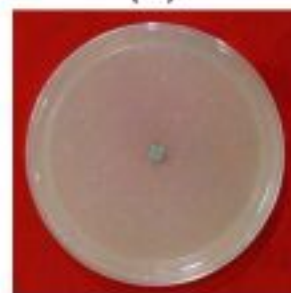
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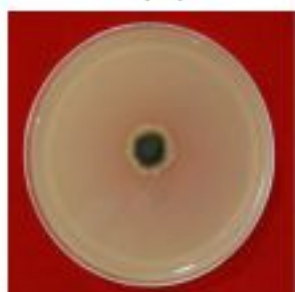
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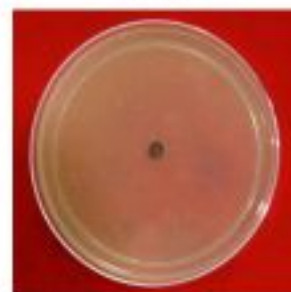
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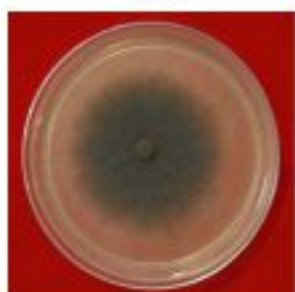
(f)



(g)



(h)



(i)



(j)

Plate 6 (III): Bio-efficacy of fungicides against *M. phaseolina* in vitro

a) Carbendazim (50% WP) @ 750 ppm

b) Sixer (75% WP) @ 1000 ppm

c) Cosko (75% SD) @ 750 ppm

d) Tilt (25% SC) @ 75 ppm

e) Amistar Top (0.125%) @ 150 ppm

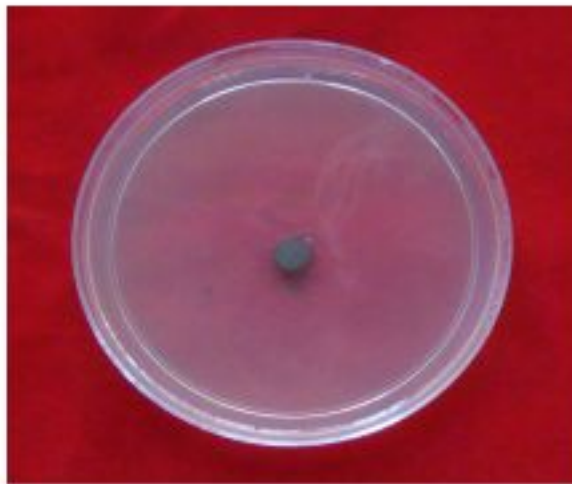
f) Thirum (75% WP) @ 1000 ppm

g) Thiophanate methyl (70% WP) @ 750 ppm

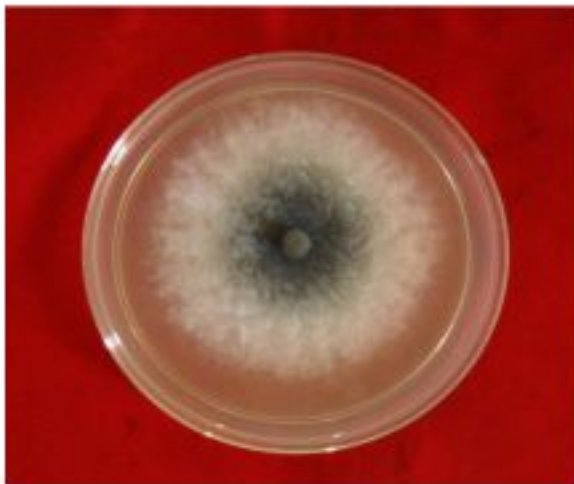
h) Vitavax (75% WP) @ 750 ppm

i) Amistar (0.1%) @ 750 ppm

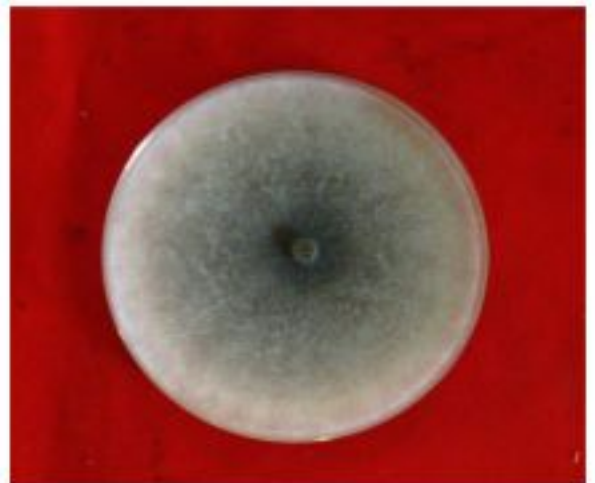
j) Control



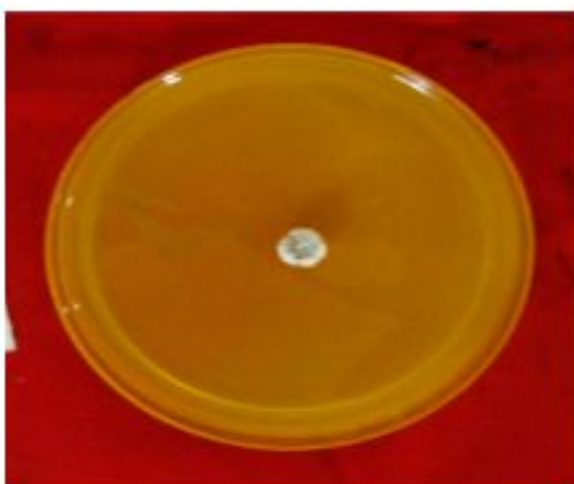
Alachlor



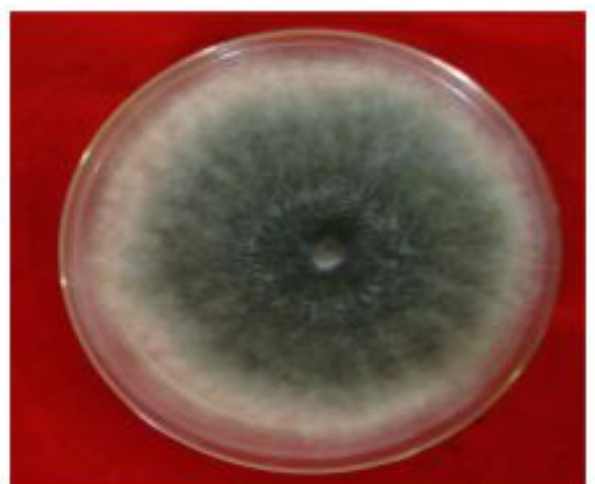
Atrazin



Metribuzin



Pendimethalin

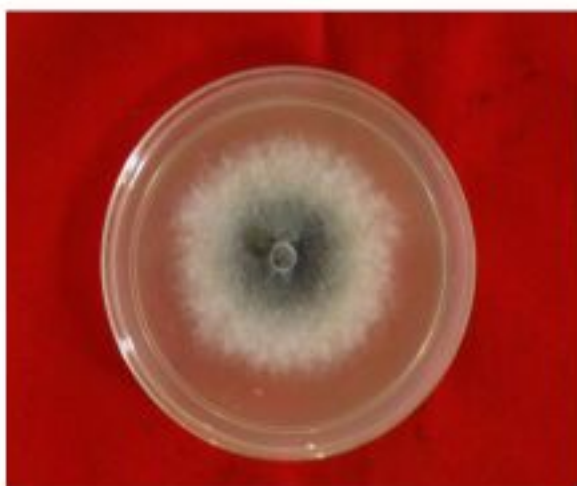


2,4- D

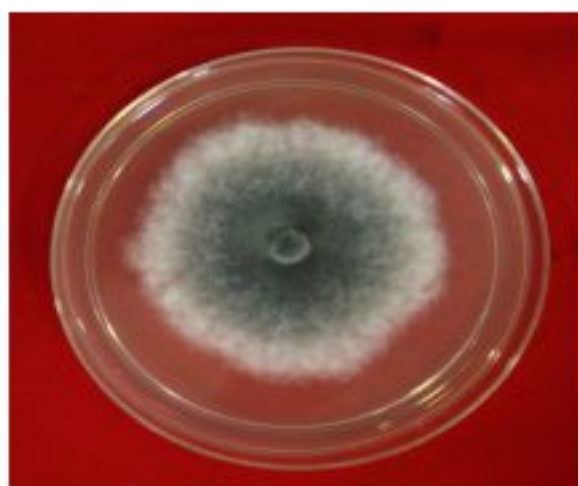
Plate 7 (I) : Bio-efficacy of herbicides (100 ppm) against
M. phaseolina in vitro



Alachlor



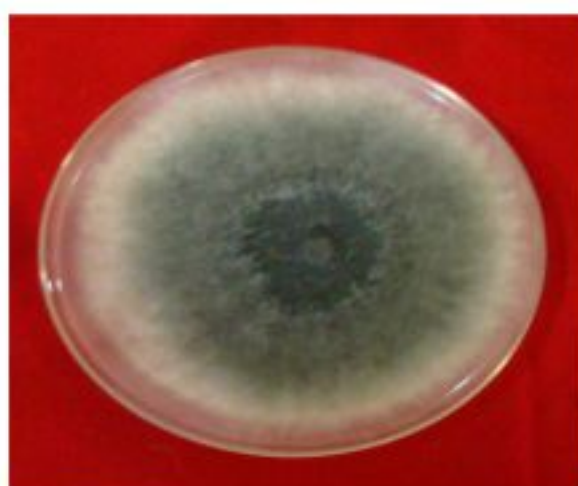
Atrazin



Metribuzin

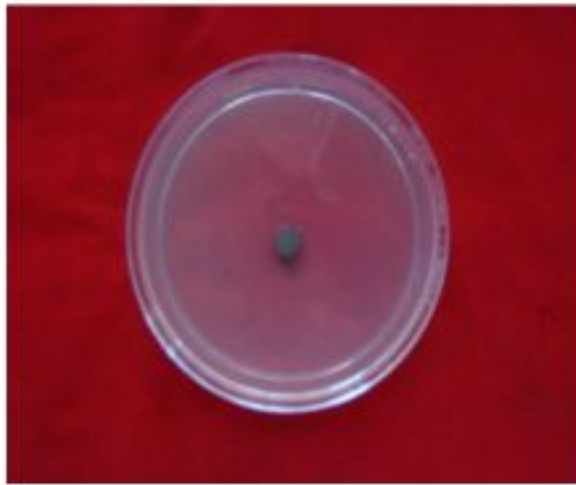


Pendimethalin

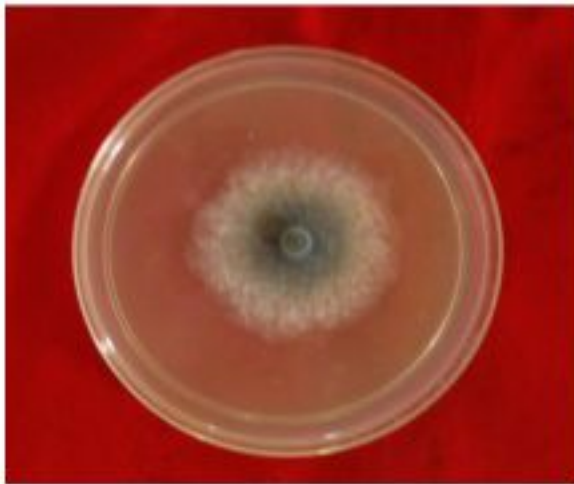


2,4- D

Plate 7 (II) : Bio-efficacy of herbicides (150 ppm) against *M. phaseolina* in vitro



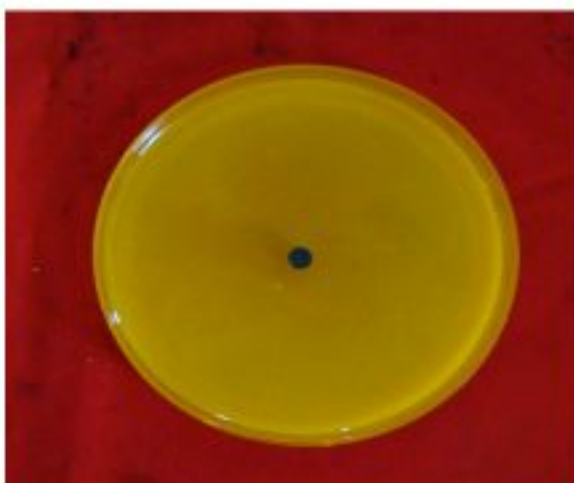
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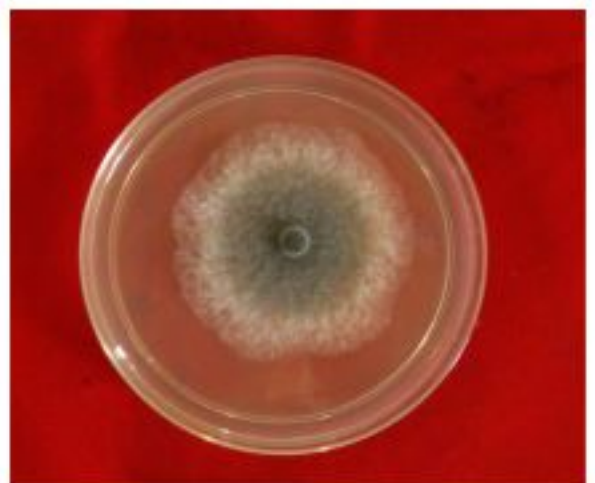
Atrazin



Metribuzin



Pendimethalin



2,4- D

Plate 7 (III) : Bio-efficacy of herbicides (200 ppm) against *M. phaseolina* *in vitro*



Uninoculated soil
(Control)



Inoculated soil



Infected Chickpea plant

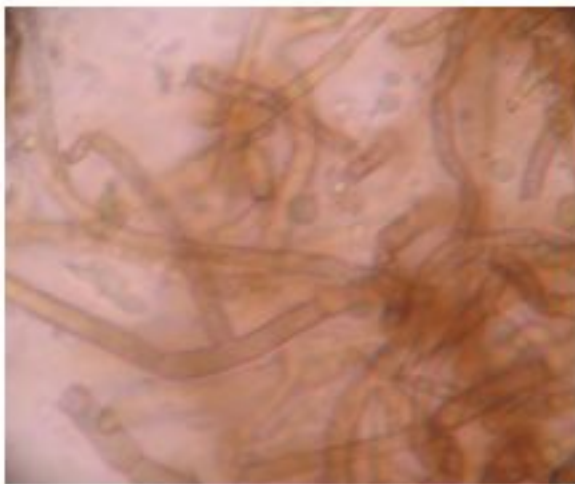


Shredding of root bark

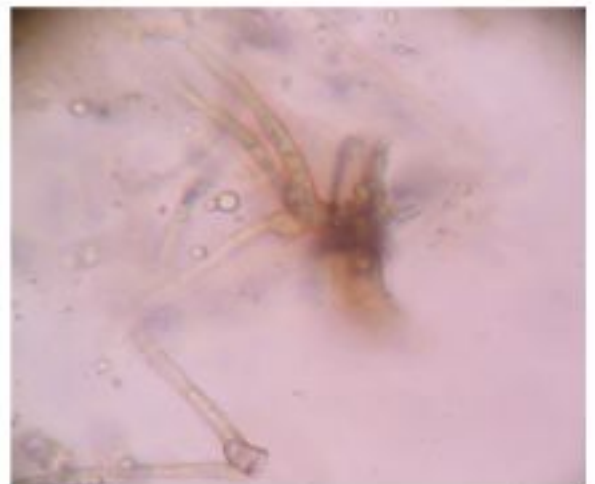
Plate 2 : Pathogenicity of *M. phaseolina* on chickpea plant



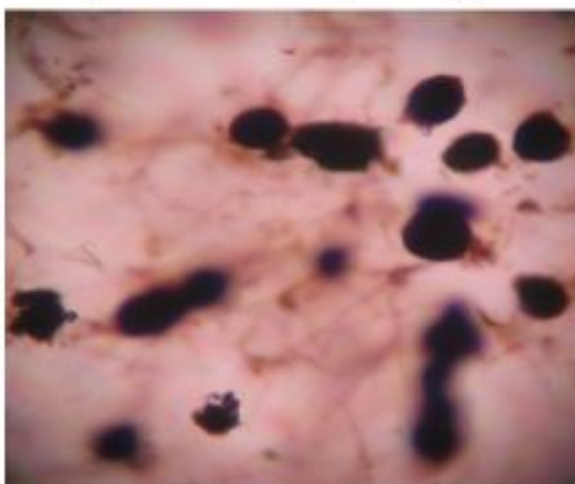
Right angle branching of mycelium



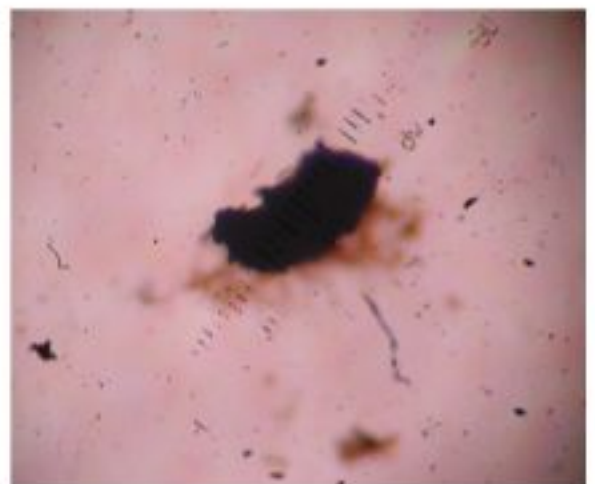
Mycelial mat of pathogen



Sclerotial formation



Well developed sclerotia



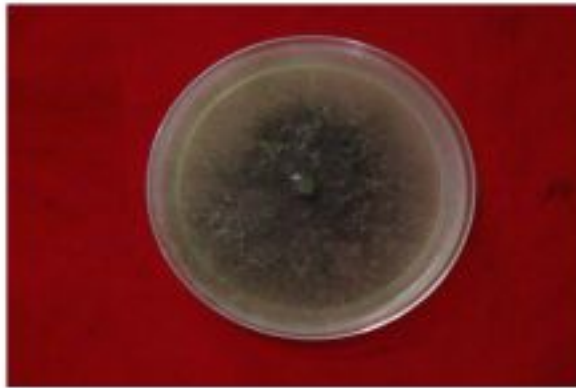
Measurement of sclerotial body

Plate 3 : Microscopy of *M. phaseolina*

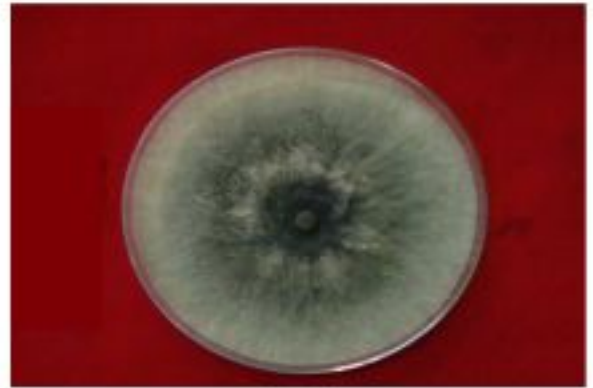


Field view of Root rot infected plants

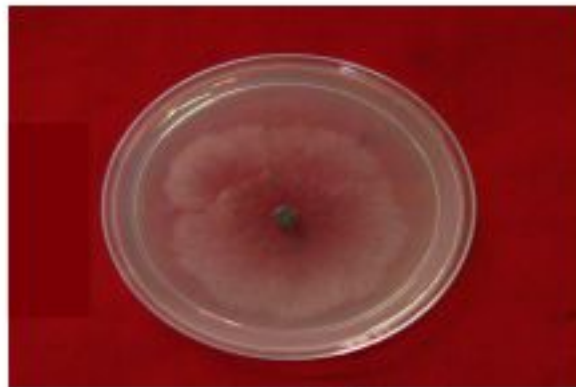
Plate 8 : Mangement of root rot of chickpea through bioagents,
oil cakes and chemicals in field conditions



CzAM



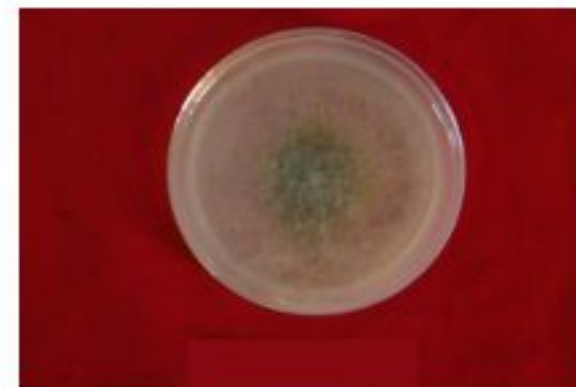
OMA



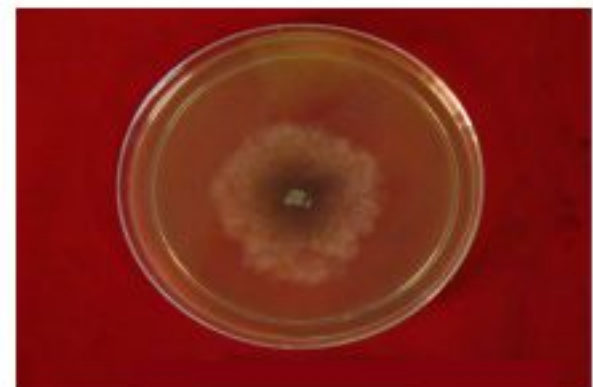
PSAM



PDA

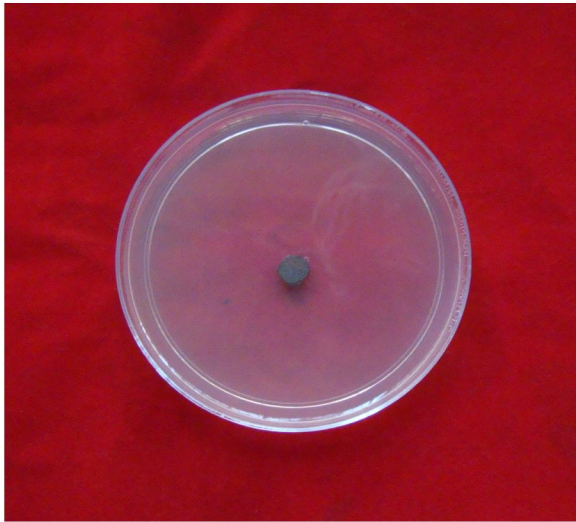


RAM

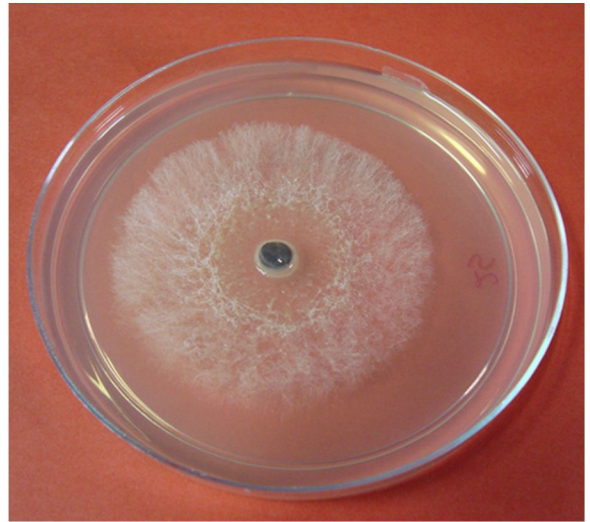


SAM

Plate 4 : Effect of solid culture media on growth of *M. phaseolina* *in vitro*



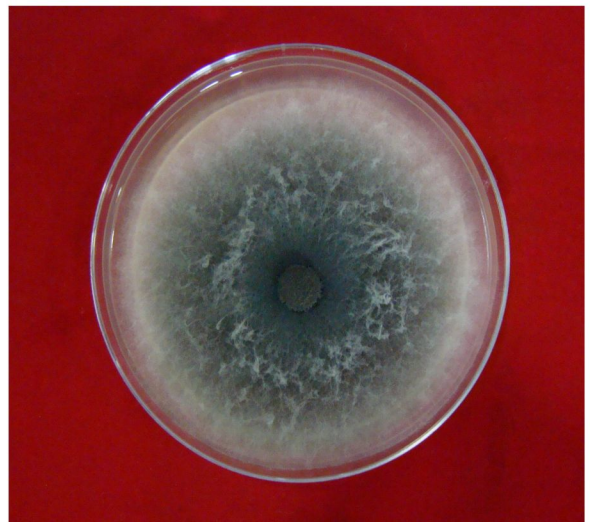
1st day of incubation



Whitish mycelial growth



Fully growth of *M. phaseolina*



Fluffy black mycelial growth
with sclerotial formation

Plate 1 : Growth pattern of *M. phaseolina* on PDA